Supplementary Materials for "`cyclinbayes`: Bayesian Causal Discovery with Linear Non-Gaussian Directed Acyclic and Cyclic Graphical Models"

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1 Background

Discovering causal relationships among variables is a fundamental task across many scientific disciplines, and causal discovery in high-dimensional settings has received growing attention in recent years. Several methods leverage the non-Gaussian error assumption, including LiNGAM [Shimizu et al., 2006] and DirectLiNGAM [Shimizu et al., 2011] for directed acyclic graphs (DAGs), with implementations available in R packages such as `pcalg` [Kalisch et al.,

2012] and `rlingam` [Kikuchi, 2025]. For directed cyclic graphs (DCGs), the LiNG-D approach [Lacerda et al., 2012] extends linear non-Gaussian causal discovery to systems containing feedback and reciprocal causal relationships.

Despite these methodological developments, software for linear non-Gaussian causal discovery remains more established for DAGs than for DCGs. Although approaches such as LiNG-D support cyclic causal discovery, available implementations remain relatively limited and generally return point estimates or candidate graph structures without quantifying uncertainty in the inferred causal relationships. When multiple graph structures are plausible given the observed data, such estimates do not reveal which causal relationships are consistently supported and which remain uncertain. A Bayesian framework provides a way to address these limitations by quantifying uncertainty in graph structures and causal effects, encouraging sparsity, and incorporating structural constraints directly into posterior inference. Although `MR.RGM` provides Bayesian inference for reciprocal causal relationships, it is designed specifically for multivariate bidirectional Mendelian randomization networks [Sarkar and Ni, 2025].

To address these limitations, we introduce `cyclinbayes`, an open-source R package for Bayesian causal discovery in both DAGs and DCGs. The proposed framework jointly learns graph structures and model parameters, using Beta–Bernoulli and spike-and-slab priors to encourage sparse graphs while allowing DAG and DCG specific structural restrictions to be incorporated directly into posterior sampling.

By sampling from the posterior distribution, `cyclinbayes` provides posterior edge inclusion probabilities, credible intervals for direct causal effects, and posterior probabilities for user-specified network motifs. These summaries allow researchers to assess uncertainty and examine the sensitivity of causal relationships across plausible graph structures. To obtain a representative graph, we further adopt a decision-theoretic framework that selects a structure by minimizing posterior expected loss under graph-discrepancy measures such

as Structural hamming Distance (SHD), Structural Intervention Distance (SID, for DAGs only), or a user-specified loss function.

Implemented in Rcpp, `cyclinbayes` uses optimized C++ routines to improve the computational efficiency of the proposed MCMC algorithms. The package is available on GitHub at `https://github.com/roblee01/cyclinbayes`.

2 Model Specification

Consider a random vector

$$\vec{\mathbf{Y}}_i = (Y_i^{(1)}, \ldots, Y_i^{(q)}, \ldots, Y_i^{(N)})^T \quad \text{for } i = 1, \ldots, p.$$

Let $Y_i^{(q)}$ be the qth realization of ith random variable or feature for $i = 1, ..., p$ and $q = 1, \ldots, N$.

Let $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ where $\mathcal{V} = \{1, \ldots, p\}$ denotes the set of nodes corresponding to the p random variables, and $\mathcal{E}$ denotes the set of directed edges representing direct causal relationships among these variables. For DAGs, the graph is acyclic, meaning that one cannot start from a node and, by following the directed edges, return to the same node. DCGs, on the other hand, allow the presence of directed cycles so that a path may begin and end at the same node. Let $pa(i)$ be the set of parents (direct causes) for node $i \in \mathcal{V}$. For a node $i \in \mathcal{V}$, let $pa(i) \subseteq \{1, \ldots, p\} \setminus \{i\}$ denote its parent set.

We consider the following linear structural causal model with mixture of Gaussians errors,

$$Y_i^{(q)} = \sum_{j \in pa(i)} B_{ij} Y_j^{(q)} + \epsilon_i^{(q)}, \tag{1}$$

where $j \in pa(i)$ means that there is a directed edge from $j \to i$, B_{ij} represents the direct causal effect from a variable $Y_j^{(q)}$ to another $Y_i^{(q)}$, and $\epsilon_i^{(q)} \sim \sum_{k=1}^{M} \pi_{ik} N(\mu_{ik}, \tau_{ik})$. We denote $\vec{\mu}_i = (\mu_{i1}, \ldots, \mu_{iM})$ and $\vec{\tau}_i = (\tau_{i1}, \ldots, \tau_{iM})$. We can write (1) in a matrix form,

$$\vec{\mathbf{Y}}^{(q)} = B\vec{\mathbf{Y}}^{(q)} + \vec{\boldsymbol{\epsilon}}^{(q)}, \tag{2}$$

where $\vec{\mathbf{Y}}^{(q)} = (Y_1^{(q)}, \ldots, Y_p^{(q)})$ and $\vec{\epsilon}^{(q)} = (\epsilon_1^{(q)}, \ldots, \epsilon_p^{(q)})^T$ are p-dimensional vectors and $B = (B_{ij})$ is an unknown $p \times p$ direct causal effect matrix subject to algorithmic constraints, including $B_{ii} = 0$ for all $i \in \{1, \ldots, p\}$.

3 Bayesian Causal Discovery Algorithms

We present Bayesian inference procedures for both DAGs and DCGs within a hierarchical framework. The goal is to estimate a sparse adjacency matrix E encoding the causal structure.

3.1 Prior Specification

Let $E = (E_{ij})_{i,j=1}^p$ denote the adjacency matrix, where $E_{ij} = 1$ indicates a directed edge $j \to i$. We impose $B_{ij} = 0$ whenever $E_{ij} = 0$, so sparsity in the graph structure directly induces sparsity in the causal effect matrix.

Adjacency matrix. Conditional on a common edge-inclusion probability γ, we assign

$$E_{ij} \mid \gamma \sim \text{Bernoulli}(\gamma), \qquad \gamma \sim \text{Beta}(a_\gamma, b_\gamma),$$

for $i \neq j$. The Beta–Bernoulli prior provides a simple mechanism for encouraging sparse graph structures while allowing the overall edge-inclusion probability to be learned from the data rather than fixed a priori. Because the inclusion probability is shared across candidate edges and assigned a Beta hyperprior, the prior also provides an adaptive form of multiplicity control as the number of possible edges increases. Further discussion of this property is provided in Appendix A.

In `BayesDAG`, E is restricted to directed acyclic graphs. In `BayesDCG`, directed cycles are permitted subject to the structural restrictions described in Section 3.4. Self loops are excluded in both models.

Causal effect matrix. Conditional on the adjacency matrix, we assign the spike and slab prior

$$B_{ij} \mid E_{ij}, \gamma_1 \sim (1 - E_{ij})\delta_0 + E_{ij} N(0, \gamma_1), \qquad \gamma_1 \sim \text{IG}(a_{\gamma_1}, b_{\gamma_1}),$$

where δ_0 denotes a point mass at zero. This prior directly links graph selection with causal effect estimation: when $E_{ij} = 0$, the corresponding causal effect is exactly zero, while included edges are allowed to have nonzero effects whose magnitude and direction are estimated from the data. The prior on γ_1 allows the scale of the nonzero effects to be learned from the data.

Together, the Beta–Bernoulli and spike and slab priors favor parsimonious graph structures while retaining flexibility in the magnitude of supported causal effects. Additional discussion of multiplicity adjustment and Bayesian complexity control is provided in Appendix A.

Mixture error distribution. For Gaussian mixture errors, we assign conjugate priors:

$$\pi_i \sim \text{Dirichlet}(\alpha, \ldots, \alpha), \qquad \mu_{ik} \sim N(a_\mu, b_\mu), \qquad \tau_{ik} \sim \text{IG}(a_\tau, b_\tau).$$

For ease of sampling, we augment the Gaussian mixture with categorical variables $Z_i^{(q)}$ such that

$$Z_i^{(q)} \mid \pi_i \sim \text{Categorical}(\pi_{i1}, \ldots, \pi_{iM}),$$

and

$$\epsilon_i^{(q)} \mid Z_i^{(q)} = k \sim N(\mu_{ik}, \tau_{ik}),$$

for $i = 1, \ldots, p$, $q = 1, \ldots, N$, and $k = 1, \ldots, M$.

3.2 DAG Structure Estimation

For DAGs, the likelihood is given by

$$p(\vec{Y}_1, \ldots, \vec{Y}_p \mid B, \boldsymbol{\mu}, \boldsymbol{\tau}, \mathbf{Z}) = \prod_{i=1}^{p} \prod_{k=1}^{M} \prod_{q=1}^{N} \left[N\left(\mu_{ik} + \sum_{j \in \text{pa}(i)} B_{ij} Y_j^{(q)},\, \tau_{ik} \right) \right]^{I(Z_i^{(q)} = k)}.$$

We integrate out B to obtain a closed form marginal likelihood for the adjacency matrix E (Appendix B.9). By averaging over the causal effect coefficients rather than conditioning on a single value, the marginal likelihood accounts for parameter uncertainty when evaluating proposed graph structures. Comparison through the marginal likelihood also provides a Bayesian Occam's-razor effect: a graph with additional edges is favored only when its improved fit is sufficient to compensate for the additional parameter space, as discussed further in Appendix A. However, direct sampling of E remains infeasible because the space of possible graph structures is enormous. Instead, we use a Metropolis–Hastings (MH) scheme with two steps for each pair (i, j):

1. (Edge addition or deletion) Propose $E_{ij} = 1$ if currently $E_{ij} = 0$ (similarly, propose $E_{ij} = 0$ if $E_{ij} = 1$). Reject the new graph if it is not acyclic. If it is acyclic, accept/reject with MH-ratio:
$$r = \frac{p(\vec{Y}_1, \ldots, \vec{Y}_p \mid E_{ij} = 1, E_{-(ij)})}{p(\vec{Y}_1, \ldots, \vec{Y}_p \mid E_{ij} = 0, E_{-(ij)})}.$$
where $E_{-(ij)} := \{E_{kl} : (k, l) \neq (i, j)\}$.

2. (Edge reversal) For an existing edge $j \to i$, propose reversing its direction to $i \to j$. Reject the proposed graph if it is not acyclic. Otherwise, accept/reject using the ratio:
$$\tilde{r} = \frac{p(\vec{Y}_1, \ldots, \vec{Y}_p \mid E_{ij} = 0, E_{ji} = 1, E_{-\{(i,j),(j,i)\}})}{p(\vec{Y}_1, \ldots, \vec{Y}_p \mid E_{ij} = 1, E_{ji} = 0, E_{-\{(i,j),(j,i)\}})}.$$

The marginalized forms of numerator and denominator of r and $\tilde{r}$ are derived in Appendix B.9.

3.3 DCG Structure Estimation

For DCGs, the lack of a recursive factorization prevents analytical marginalization of B. We therefore jointly sample (E, B). For an edge addition, we propose $E_{ij} = 1$ together with $B_{ij} \sim N(0, \sigma^2_{\text{add}})$ and compute the corresponding Metropolis–Hastings ratio (Appendix C).

For existing edges, we use random walk updates

$$B'_{ij} \sim N(B_{ij}, \sigma^2_{\text{rw}}).$$

Posterior inference is based on samples from the admissible DCG space described in Section 3.4.

During burn-in, `BayesDCG` is allowed to explore a broader unconstrained DCG space without imposing either the disjoint cycle or stability restrictions. These samples are discarded and are used only for exploration and initialization of the constrained phase. At the end of burn in, the current state is projected into the stable disjoint cycle space using the procedure described in Appendix C.4. Thereafter, graph proposals that violate the disjoint cycle restriction or the stability condition $\rho(B) < 1$ are rejected. Only samples from this constrained phase are retained for posterior inference.

3.4 Identifiability, Stability, and Causal Interpretation

Model assumptions. The DAG and DCG models are considered under the linear non-Gaussian structural equation framework introduced in Section 2. Both models rely on the following assumptions:

(A1) *Linearity.* The relationships among the observed variables are described by linear structural equations.

(A2) *Independent observations.* The observations are assumed to be independently and identically distributed.

(A3) *Independent non-Gaussian structural errors.* The structural errors are mutually independent across variables, non-Gaussian, and have nonzero variance.

(A4) *Causal sufficiency.* There are no unobserved common causes among the modeled variables.

Under these assumptions, non-Gaussianity provides information beyond second-order dependence that can help distinguish causal structures which would not generally be identifiable under Gaussian errors alone. However, the additional structural conditions required for identifiability differ between the acyclic and cyclic settings, as discussed below.

Identifiability. For the acyclic setting, `BayesDAG` follows the standard LiNGAM framework, under which the causal ordering and direct causal effect matrix B in (2) are identifiable from the observational distribution [Shimizu et al., 2006]. The assumptions stated above, together with acyclicity, therefore provide the identifiability conditions for `BayesDAG`.

The cyclic setting, `BayesDCG` requires additional structure. Unlike the acyclic case, non-Gaussianity alone does not generally guarantee unique identification of a cyclic SEM, because different cyclic structures may generate the same observational distribution. We therefore consider DCGs with disjoint directed cycles, meaning that no two directed cycles share a node, together with the stability condition described below.

The identifiability argument of Lacerda et al. [2012] motivates this restriction. For a directed cycle C, consider the product of its edge coefficients,

$$\pi_C = \prod_{(j \to i) \in C} B_{ij}.$$

When the original cycle satisfies $|\pi_C| < 1$, the corresponding reversed cycle has product magnitude greater than one and is unstable. For disjoint cycles, this argument applies separately to each cycle, allowing stability to distinguish the unique stable SEM within the relevant distribution-equivalence class.

Importantly, the disjoint cycle condition is sufficient rather than necessary for this identifiability result. More general cyclic graphs may also be identifiable, but uniqueness is not guaranteed under the assumptions considered here. Accordingly, `BayesDCG` restricts retained posterior inference to stable DCGs with disjoint cycles.

Stability. The structural model in (2) can be rearranged as

$$(I - B)\vec{\mathbf{Y}}^{(q)} = \vec{\boldsymbol{\epsilon}}^{(q)}.$$

When $I - B$ is invertible, the model admits the reduced-form representation

$$\vec{\mathbf{Y}}^{(q)} = (I - B)^{-1}\vec{\boldsymbol{\epsilon}}^{(q)}.$$

For DAGs, invertibility follows automatically from acyclicity. Every DAG admits a topological ordering under which the direct causal effect matrix B is strictly triangular. Consequently, all eigenvalues of B are zero, so $\rho(B) = 0 < 1$ and no separate stability restriction is required.

For DCGs, invertibility and stability are no longer automatic because directed feedback is permitted. We therefore impose the spectral-radius condition

$$\rho(B) = \max_{\lambda \in \text{eig}(B)} |\lambda| < 1.$$

This condition guarantees that $I - B$ is invertible and that perturbations propagated repeatedly through feedback loops decay rather than amplify indefinitely. Accordingly, proposed states violating $\rho(B) < 1$ are rejected during the constrained posterior sampling phase of `BayesDCG`.

Thus, in the cyclic setting, stability ensures a well-defined feedback system and, together with the disjoint cycle restriction, identifies the unique stable representative within the relevant distribution-equivalence class.

Causal interpretation. Under the assumptions above, identifiability determines whether the structural coefficient matrix B can be uniquely recovered, while stability ensures that the cyclic system is well defined. Conditional on these assumptions, the entries of B retain their structural causal interpretation. Specifically, as in Section 2, B_{ij} represents the direct causal effect associated with the edge $j \to i$.

In a DAG, indirect effects may propagate along directed paths, but feedback is excluded by acyclicity. In a DCG, directed feedback is permitted, so a perturbation may propagate repeatedly through a feedback loop. Consequently, an individual coefficient B_{ij} should be interpreted as a direct structural effect rather than as the total effect of an intervention after accounting for all feedback pathways.

4 Algorithms

The MCMC algorithms for DAGs and DCGs are provided in Algorithms 1 and 2, respectively. Full conditional distributions are provided in Appendices B and C.

The main computational distinction between the two algorithms lies in their treatment of the graph structure and causal effect coefficients. For `BayesDAG`, we marginalize over B to obtain a closed form marginal likelihood for graph updates (Appendix B.9). In contrast, `BayesDCG` jointly samples E and B through Metropolis–Hastings updates, subject to the structural restrictions described in Section 3.4.

Algorithm 1 `BayesDAG`: Bayesian Sparse Collapsed LiNGAM for DAGs

Initialize $\gamma, \gamma_1, E, B, \{\pi_i, \boldsymbol{\mu}_i, \boldsymbol{\tau}_i, \mathbf{Z}_i\}_{i=1}^p$
for $m = 1, \ldots, M_{\text{iter}}$ **do**
  Sample $\gamma \mid E$ ▷ Appendix B
  for $i = 1, \ldots, p$ **do**
    Sample $\tau_{ik} \mid \epsilon_i, Z_i,\ k = 1, \ldots, M$ ▷ Appendix B
  end for
  for each (i, j), $i \neq j$ **do**
    Compute collapsed edge ratio $r = p(Y \mid E_{ij} = 1, E_{-(ij)})/p(Y \mid E_{ij} = 0, E_{-(ij)})$ ▷ Appendix B.9
    Set $r_m = r(c_2 r)^{\eta(m)}$ during annealing ▷ Section B.10
    Accept edge addition/deletion with probability $\min(1, r_m)$ if acyclic
    if $E_{ij} = 1$ **then**
      Compute reversal ratio $\tilde{r}_m$ and accept $j \to i \leftrightarrow i \to j$ if acyclic ▷ Appendix B.9
    end if
  end for
  for $i = 1, \ldots, p$ **do**
    Sample $\mu_{ik} \mid \epsilon_i, Z_i, \tau_{ik},\ k = 1, \ldots, M$ ▷ Appendix B
  end for
  Sample $B \mid E, Y, Z$ ▷ Appendix B
  Compute $\epsilon = (I - B)Y$
  Sample $\gamma_1 \mid B, E$ ▷ Appendix B
  for $i = 1, \ldots, p$ **do**
    Sample $Z_i \mid \epsilon_i, \pi_i, \mu_i, \tau_i$ and $\pi_i \mid Z_i$ ▷ Appendix B
  end for
end for

Algorithm 2 `BayesDCG`: Bayesian Sparse Causal Discovery for DCGs

Initialize $\gamma, \gamma_1, E, B, \{\pi_i, \mu_i, \tau_i, Z_i\}_{i=1}^{p}$
for $m = 1, \ldots, M_{\text{iter}}$ **do**
  Sample γ, γ_1 and mixture parameters as in Algorithm 1 ▷ Appendix B
  if $m = M_{\text{burn}} + 1$ **then**
    Project (E, B) into the disjoint cycle space ▷ Algorithm 3
  end if
  for each (i, j), $i \neq j$ **do**
    Propose add/delete of E_{ij}; if adding, $B'_{ij} \sim N(0, \sigma^2_{\text{add}})$ ▷ Appendix C
    if $\rho(B') \geq 1$ **then**
      Reject ▷ Section 3.4
    else if $m > M_{\text{burn}}$ and E' violates disjoint cycles **then**
      Reject ▷ Section 3.4
    else
      Accept/reject (E', B') using the MH ratio ▷ Appendix C
    end if
  end for
  for each (i, j) with $E_{ij} = 1$ **do**
    Propose $B'_{ij} \sim N(B_{ij}, \sigma^2_{\text{rw}})$
    if $\rho(B') < 1$ **then**
      Accept/reject using the MH ratio ▷ Appendix C
    else
      Reject
    end if
  end for
end for

4.1 Posterior Graph Selection

Both algorithms produce posterior samples of graph structures $\{\mathcal{G}^{(1)}, \ldots, \mathcal{G}^{(m)}\}$, allowing uncertainty over causal structures to be quantified. While edge wise posterior inclusion

probabilities provide useful summaries, independently thresholding these probabilities may produce a graph that violates the structural restrictions of the fitted model, such as acyclicity or the disjoint cycle restriction.

To obtain a representative graph while respecting these constraints, we formulate graph selection as a Bayesian decision problem. Let $d(\cdot, \cdot)$ denote a graph discrepancy measure. Here, $a \in \mathcal{D}$ denotes a candidate graph to be selected, $g \in \mathcal{D}$ denotes a possible graph under the posterior, and $\pi(g \mid \vec{Y})$ denotes the posterior probability assigned to graph g given the observed data $\vec{Y}$. The Bayes-optimal graph minimizes posterior expected loss:

$$\hat{\mathcal{G}} = \arg\min_{a \in \mathcal{D}} \sum_{g \in \mathcal{D}} d(a, g)\, \pi(g \mid \vec{Y}),$$

where $\mathcal{D}$ denotes the admissible graph space under the fitted model.

We approximate this decision rule using the posterior graph samples. Let $\{\mathcal{G}_*^{(1)}, \ldots, \mathcal{G}_*^{(v)}\}$ denote the unique sampled graphs with corresponding frequencies $w_1, \ldots, w_v$. We select the posterior weighted medoid:

$$\hat{\mathcal{G}} = \arg\min_{l \in \{1, \ldots, v\}} \sum_{u=1}^{v} w_u\, d(\mathcal{G}_*^{(l)}, \mathcal{G}_*^{(u)}).$$

For SHD, this decision rule preserves posterior graph consistency: if the posterior probability of the true graph $\mathcal{G}_0$ converges to one, then the Bayes action minimizing posterior expected SHD converges to $\mathcal{G}_0$. A short proof is provided in Appendix D.

The package supports Structural Hamming Distance (SHD), Structural Intervention Distance (SID, for DAGs only), and user-defined graph discrepancy measures. Our theoretical consistency discussion focuses on SHD. SID is included as an alternative decision-theoretic loss for DAGs, but we do not claim a separate consistency result for the SID based summary.

5 Design and Implementation

The package `cyclinbayes` provides two MCMC samplers and three summary functions, all implemented in Rcpp for efficiency.

Samplers

- `BayesDAG()`: Bayesian LiNGAM sampler using collapsed Gibbs updates for the graph structure with simulated annealing. Returns posterior draws of the adjacency matrix E, causal effect matrix B, and mixture parameters.

- `BayesDCG()`: Bayesian sampler for directed cyclic graphs. Jointly updates (E, B) via Metropolis–Hastings with random-walk proposals. Output format matches `BayesDAG()`.

Summary Functions

- `posterior_interval_est()`: Computes HPD and equal-tailed credible intervals for model parameters.

- `point_est_graph()`: Returns a point estimate of the graph by selecting the posterior weighted medoid under a specified distance (SHD, SID, or custom).

- `posterior_network_motif()`: Computes the posterior probability that a target subgraph $\mathcal{G}$ is contained in the sampled graphs:

$$\hat{\pi} = \frac{1}{m} \sum_{t=1}^{m} I(\mathcal{G} \subseteq \mathcal{G}^{(\mathrm{t})}).$$

Comparison with Existing Software

Table 1 compares `cyclinbayes` with existing causal discovery implementations in terms of their inferential capabilities, computational requirements, and software availability.

`ICA-LiNGAM` and `DirectLiNGAM` estimate causal structures for DAGs, while `LiNG-D` allows directed cycles. These methods provide estimated graphs without posterior uncertainty quantification. `MR.RGM` offers Bayesian inference for reciprocal causal relationships but does not directly provide credible intervals for individual causal effect coefficients in its standard output. In comparison, `cyclinbayes` provides Bayesian inference for both DAGs and

DCGs, including uncertainty quantification for graph structures and causal effects, posterior network motif probabilities, and decision-theoretic graph selection.

Software / method	DAG	DCG	Non-Gauss.	Posterior uncertainty	Effect intervals	Graph-level summaries	Runtime	Peak memory	Distribution	Status
ICA-LiNGAM	✓	–	✓	–	–	–	Fast	Low	GitHub	Public
DirectLiNGAM	✓	–	✓	–	–	–	Fast	Low	GitHub	Public
`LiNG-D`	–	✓	✓	–	–	–	Fast	Low	GitHub	Public
`MR.RGM`	–	✓	–	✓	–	✓	Moderate	Moderate	CRAN	Maintained
`cyclinbayes` (`BayesDAG`/`BayesDCG`)	✓	✓	✓	✓	✓	✓	Longer	High	GitHub	Current

Table 1: Comparison of methodological and software features among causal discovery implementations.

Notes. Results are based on the setting $N = 100$, $p = 20$, with Laplace errors. Runtime is averaged over 100 simulations, while peak memory is measured from one representative fit. Runtime categories are defined as Fast (0.09–1.36 s), Moderate ($\approx$ 27 s), and Longer (223–362 s). Memory categories are Low (118–301 MB), Moderate ($\approx$ 442 MB), and High (633–643 MB). `cyclinbayes` includes both `BayesDAG` and `BayesDCG`; `BayesDCG` can also accommodate DAGs. All runtime and memory categories refer to this representative setting.

The comparison highlights the additional computational cost of Bayesian posterior sampling. Under the representative setting, `ICA-LiNGAM`, `DirectLiNGAM`, and `LiNG-D` require less than two seconds, while `MR.RGM` requires approximately 27 seconds. In comparison, `BayesDAG` and `BayesDCG` require approximately 362 and 223 seconds, respectively. Their peak memory usage is approximately 633 and 643 MB, compared with 118–301 MB for the point-estimation procedures and 442 MB for `MR.RGM`. These results illustrate the statistical–computational tradeoff of `cyclinbayes`: posterior inference provides richer and better summaries of graph structures and causal effects but requires additional computation. Detailed graph recovery and runtime comparisons are provided in Section 6.5.

6 Examples

This section illustrates the primary functions in `cyclinbayes` for fitting acyclic and cyclic causal models, selecting representative posterior graphs, evaluating posterior support for net-

work structures, and summarizing posterior uncertainty. Complete reproducible examples, including data generation and initialization, are provided in the package README.

6.1 Acyclic Example (DAG)

6.1.1 Running `BayesDAG()`

Given an $N \times p$ data matrix , `BayesDAG()` performs posterior inference over DAG structures, causal effect coefficients, sparsity parameters, and Gaussian mixture error parameters. A typical call is

```
results_list = BayesDAG(
  data_matrix,
  params$a_mu, params$b_mu,
  params$a_gamma, params$b_gamma,
  params$a_tao, params$b_tao,
  params$a_og_tao, params$b_og_tao,
  params$a_gamma_1, params$b_gamma_1,
  params$alpha,
  M,
  num_iter
)
```

Posterior samples can then be extracted from the returned object:

```
Adjacency_matrix_list =
  results_list$Adjacency_matrix_list


Causal_effect_matrix_list =
  results_list$Causal_effect_matrix_list
```

```
log_likelihood_list =
  results_list$log_likelihood_list


gamma_list =
  results_list$gamma_list


gamma_1_list =
  results_list$gamma_1_list


mu_matrix_list =
  results_list$mu_matrix_list


tao_matrix_list =
  results_list$tao_matrix_list


pi_matrix_list =
  results_list$pi_matrix_list
```

Matrix valued posterior draws are stored in vectorized form, with rows corresponding to posterior iterations and columns corresponding to fixed parameter entries.

6.1.2 Selecting a Posterior Graph

A representative posterior graph can be obtained using `point_est_graph()`. Our primary graph loss is structural hamming distance (SHD):

```
Adjacency_est_shd = point_est_graph(
  Adjacency_matrix_list,
  dist_type = "shd"
)
```

For DAGs, structural intervention distance (SID) is also available as a secondary loss:

```
Adjacency_est_sid = point_est_graph(
  Adjacency_matrix_list,
  dist_type = "sid"
)
```

Users may also supply a custom graph distance:

```
custom_edge_mismatch = function(A, B) {
  sum(abs(A - B))
}

Adjacency_est_custom = point_est_graph(
  Adjacency_matrix_list,
  dist_type = "custom",
  dist_fun = custom_edge_mismatch
)
```

This allows users to examine the sensitivity of the selected graph to the choice of graph loss.

6.1.3 Posterior Support for Network Structures

The function `posterior_network_motif()` evaluates posterior support for a specified network structure or motif. If a candidate structure is represented by an adjacency matrix `Adjacency_candidate`, its posterior support can be evaluated using

```
candidate_graph =
  igraph::graph_from_adjacency_matrix(
    Adjacency_candidate
  )
```

```
posterior_network_motif(
  candidate_graph,
  Adjacency_matrix_list
)
```

The returned value is the proportion of posterior graph samples in which all edges in the specified motif are present.

6.1.4 Posterior Interval Estimates

Posterior uncertainty in model parameters can be summarized using `posterior_interval_est()`. For example, intervals for the causal effect coefficients are obtained by

```
Causal_effect_summary =
  posterior_interval_est(
    Causal_effect_matrix_list,
    level = 0.95
  )

hpd_matrix =
  Causal_effect_summary$hpd_matrix

ci_matrix =
  Causal_effect_summary$ci_matrix
```

The function returns highest posterior density (HPD) intervals and equal-tailed credible intervals for each entry of the sampled causal effect matrix. The same function can also be applied to other sampled model parameters returned by `BayesDAG()`.

6.1.5 Inspecting Sampler Behavior

The log-likelihood trajectory may be inspected as an empirical diagnostic of sampler stabilization:

```
plot(
  log_likelihood_list,
  type = "l",
  xlab = "Iteration",
  ylab = "Log-likelihood"
)
```

6.2 Cyclic Example (DCG)

The workflow for DCGs parallels the acyclic case. After running the sampler, users can select a representative posterior graph, quantify posterior support for network structures, summarize uncertainty in causal effect coefficients, and inspect the log-likelihood trajectory. The main distinction is that `BayesDCG()` permits feedback relationships and performs posterior inference within the cyclic model space.

6.2.1 Running `BayesDCG()`

Given an $N \times p$ data matrix `data_matrix`, `BayesDCG()` can be called as follows:

```
results_list = BayesDCG(
  data_matrix = data_matrix,
  a_mu      = params$a_mu,
  b_mu      = params$b_mu,
  a_gamma   = params$a_gamma,
  b_gamma   = params$b_gamma,
  a_tao     = params$a_tao,
```

```
  b_tao      = params$b_tao,
  a_gamma_1  = params$a_gamma_1,
  b_gamma_1  = params$b_gamma_1,
  alpha      = params$alpha,
  M          = M,
  num_iter   = num_iter,
  burn_in_iterations = burn_in_iterations
)
```

The returned object contains posterior draws of the same principal model parameters as `BayesDAG()`, including adjacency matrices, causal effect matrices, mixture parameters, and sparsity parameters. For example,

```
Adjacency_matrix_list =
  results_list$Adjacency_matrix_list


Causal_effect_matrix_list =
  results_list$Causal_effect_matrix_list


log_likelihood_list =
  results_list$log_likelihood_list
```

The argument `burn_in_iterations` specifies the initial exploration period before posterior sampling proceeds in the constrained cyclic model space.

6.2.2 Posterior Analysis and Uncertainty Quantification

Posterior analysis for `BayesDCG()` follows the same workflow as in the acyclic example. A representative graph can be selected using `point_est_graph()`, posterior support for a specified network structure can be evaluated using `posterior_network_motif()`, and pos-

terior intervals for causal effect coefficients or other model parameters can be obtained using `posterior_interval_est()`.

For example, an SHD-based representative graph is obtained using

```
Adjacency_est_shd = point_est_graph(
  Adjacency_matrix_list,
  dist_type = "shd"
)
```

and posterior intervals for the causal effect matrix are obtained using

```
Causal_effect_summary =
  posterior_interval_est(
    Causal_effect_matrix_list,
    level = 0.95
  )

hpd_matrix =
  Causal_effect_summary$hpd_matrix

ci_matrix =
  Causal_effect_summary$ci_matrix
```

Posterior support for a candidate cyclic network structure is computed in the same manner as in the DAG example using `posterior_network_motif()`.

The principal distinction in posterior graph selection is that structural intervention distance (SID) is defined for acyclic graphs. Therefore, SHD or a user-defined distance should be used when the posterior sample contains cyclic graph structures.

As in the DAG case, the log-likelihood trajectory can also be inspected as an empirical diagnostic of sampler stabilization:

```
plot(
  log_likelihood_list,
  type = "l",
  xlab = "Iteration",
  ylab = "Log-likelihood"
)
```

6.3 Simulation Settings

We evaluate `BayesDAG` and `BayesDCG` across sample sizes $N \in \{100, 200, 300\}$, graph dimensions $p \in \{20, 30, 40\}$, and three structural error distributions. For each setting, we generate 100 independent data sets while holding the underlying graph structure and causal effect matrix fixed across replications.

The structural errors are independently generated across variables and observations from one of the following distribution cases,

(C1) A two-component Gaussian mixture:

$$\epsilon_i^{(q)} \sim 0.5N(-0.5, 0.1^2) + 0.5N(0.5, 0.3^2).$$

(C2) A Laplace distribution:

$$\epsilon_i^{(q)} \sim \text{Laplace}(0, 0.25).$$

(C3) A Student's t distribution:

$$\epsilon_i^{(q)} \sim t_7.$$

Both Bayesian methods are fitted using a five-component Gaussian mixture error model ($M = 5$). The Laplace and Student's t settings therefore also assess robustness to misspecification of the structural error distribution.

For the acyclic experiments, we generate sparse DAGs with edge probability 0.10. For the cyclic experiments, we generate DCGs under the structural restrictions imposed by `BayesDCG`. In both settings, nonzero causal effect coefficients are generated according to

$$|B_{ij}| \sim \text{Unif}(0.4, 0.9),$$

with independent positive or negative signs chosen with equal probability. The observed data are then generated from

$$\mathbf{Y}^{(q)} = (I - B)^{-1}\boldsymbol{\epsilon}^{(q)}, \qquad q = 1, \ldots, N.$$

For the acyclic experiments, we compare `BayesDAG` with `ICA-LiNGAM` [Shimizu et al., 2006] and `DirectLiNGAM` [Shimizu et al., 2011]. For the cyclic experiments, we compare `BayesDCG` with `LiNG` [Lacerda et al., 2012] and `MR.RGM` [Sarkar and Ni, 2025]. Performance is evaluated using TPR, FPR, accuracy, precision, recall, F1 score, and computational runtime.

Both Bayesian procedures are run for 100,000 MCMC iterations. For `BayesDAG`, the initial portion of the chain is discarded as burn-in. For `BayesDCG`, the sampler initially explores the unrestricted cyclic space before transitioning into the admissible constrained space used for posterior inference. Further details on the sampling schedules and convergence diagnostics are provided in Section 6.5.

6.4 Evaluation Metrics

For each of our simulation settings varied by (N, p) and the distribution of error ϵ, we access the precision of our corresponding adjacency matrix the true positive rate (TPR), the false positive rate (FPR), precision, accuracy, recall, and F1 score of E, which are defined as

follows:

$$
\begin{aligned}
TPR &= \frac{\sum I(E_{ij}=1,\hat{E}_{ij}=1)}{\sum I(E_{ij}=1,\hat{E}_{ij}=1)+\sum I(E_{ij}=1,\hat{E_{ij}}=0)};\\
FPR &= \frac{\sum I(E_{ij}=0,\hat{E}_{ij}=1)}{\sum I(E_{ij}=0,\hat{E}_{ij}=1)+\sum I(E_{ij}=0,\hat{E}_{ij}=0)};\\
Precision &= \frac{\sum I(E_{ij}=1,\hat{E}_{ij}=1)}{\sum I(E_{ij}=1,\hat{E}_{ij}=1)+\sum I(E_{ij}=0,\hat{E}_{ij}=1)};\\
Recall &= \frac{\sum I(E_{ij}=1,\hat{E}_{ij}=1)}{\sum I(E_{ij}=1,\hat{E}_{ij}=1)+\sum I(E_{ij}=1,\hat{E}_{ij}=0)};\\
Accuracy &= \frac{\sum I(E_{ij}=1,\hat{E}_{ij}=1)+\sum I(E_{ij}=0,\hat{E_{ij}}=0)}{\sum I(E_{ij}=1,\hat{E}_{ij}=1)+\sum I(E_{ij}=1,\hat{E}_{ij}=0)+\sum I(E_{ij}=0,\hat{E_{ij}}=0)+\sum I(E_{ij}=0,\hat{E}_{ij}=1)};\\
F1 &= 2\frac{Precision\cdot Recall}{Precision+Recall},
\end{aligned}
$$

where $\hat{E}_{ij}$ and E_{ij} represent the estimated adjacency matrix and the true adjacency matrix, respectively.

6.5 Simulation Results

Acyclic Results. As shown in Tables 2–4, `BayesDAG` demonstrates strong graph recovery performance across the considered acyclic simulation settings, with its advantage particularly apparent at $N = 100$. Compared with `ICA-LiNGAM` and `DirectLiNGAM`, the proposed method generally recovers a larger proportion of the true edges while maintaining higher precision and lower false positive rates. This pattern becomes especially important as graph dimension increases, when the number of possible directed edges grows substantially. For example, under Gaussian-mixture errors with $N = 100$ and $p = 40$, `BayesDAG` attains an F1 score of 0.9245, compared with 0.3895 for `ICA-LiNGAM` and 0.5525 for `DirectLiNGAM`.

The results also demonstrate the flexibility of the finite Gaussian-mixture error model when the generating distribution is misspecified. Under Laplace errors with $N = 100$ and $p = 40$, the F1 scores are 0.8995 for `BayesDAG`, 0.3621 for `ICA-LiNGAM`, and 0.4418 for `DirectLiNGAM`. A similar pattern is observed under Student's t errors with $N = 100$ and

$p = 30$, where `BayesDAG`, `ICA-LiNGAM`, and `DirectLiNGAM` achieve F1 scores of 0.7235, 0.3786, and 0.4412, respectively.

As the sample size increases, however, `DirectLiNGAM` becomes increasingly competitive. For instance, under Laplace errors with $N = 300$, it attains higher F1 scores than `BayesDAG` at $p = 20$ and $p = 30$, while `BayesDAG` retains a slight advantage at $p = 40$. Together, these findings suggest that the benefits of the Bayesian approach are particularly pronounced in smaller sample settings, while its mixture formulation remains effective across the considered non-Gaussian error distributions. The observed false-positive control is also consistent with the sparsity and complexity control induced by the Beta–Bernoulli graph prior discussed in Appendix A.

These statistical benefits come with additional computational cost. Under Laplace errors with $N = 100$, the mean runtime of `BayesDAG` increases from approximately 362 seconds at $p = 20$ to 831 seconds at $p = 30$ and 1828 seconds at $p = 40$. In comparison, `DirectLiNGAM` requires approximately 1.4, 2.8, and 5.6 seconds, respectively, while `ICA-LiNGAM` requires less than one second in each setting. Thus, the results highlight a statistical–computational tradeoff: `BayesDAG` provides strong graph recovery in the considered settings while additionally supporting posterior inference over graph structures and causal effects, but requires substantially greater computation than the point estimation procedures.

Cyclic Results. As shown in Tables 5–7, `BayesDCG` achieves substantially better graph recovery performance than both the `LiNG` and `MR.RGM` baselines across the cyclic simulation settings. Although `LiNG` and `MR.RGM` attain very high recall, they also introduce many additional edges, resulting in high false-positive rates and low precision. In contrast, `BayesDCG` maintains much lower false-positive rates while preserving relatively high recall, leading to substantially higher precision, accuracy, and F1 scores. The comparison with `LiNG` is particularly informative because both methods target linear non-Gaussian cyclic models. As the sample size increases, the recall of `BayesDCG` generally improves while its false-positive rate

remains low.

This pattern is observed across the considered Gaussian-mixture, Laplace, and t error settings. For example, under Laplace errors with $N = 100$ and $p = 40$, `BayesDCG` attains an F1 score of 0.6880, compared with 0.1900 for `LiNG` and 0.1821 for `MR.RGM`. Its false-positive rate is also substantially lower, 0.0425 compared with 0.8904 and 0.9658, respectively. As the number of possible directed edges increases with graph dimension, controlling spurious edge inclusion becomes increasingly important. The observed pattern is consistent with the sparsity and complexity control induced by the Beta–Bernoulli graph prior discussed in Appendix A, together with the structural restrictions imposed by the cyclic sampler.

These improvements come with additional computational cost. Under Laplace errors with $N = 100$, the mean runtime of `BayesDCG` increases from approximately 223 seconds at $p = 20$ to 725 seconds at $p = 30$ and 3849 seconds at $p = 40$. The corresponding runtimes for `MR.RGM` are approximately 27, 89, and 213 seconds, while `LiNG` requires less than one second in each setting. Similar computational patterns are observed under Gaussian-mixture and t errors. Thus, the results demonstrate a statistical–computational tradeoff: `BayesDCG` provides improved graph recovery and posterior inference over graph structures and causal effects, but requires greater computation than the comparison methods, particularly as graph dimension increases.

Convergence of `BayesDAG`. Figure 1 presents the log-likelihood trajectory for a representative `BayesDAG` simulation with Laplace errors, $p = 20$, and $N = 100$. The sampler uses a gradually decaying annealing schedule, with the vertical dashed line marking the point at which the annealing contribution becomes negligible. After the initial transition, the log-likelihood fluctuates around a relatively stable level, providing empirical evidence that the sampler has entered a stable sampling regime.

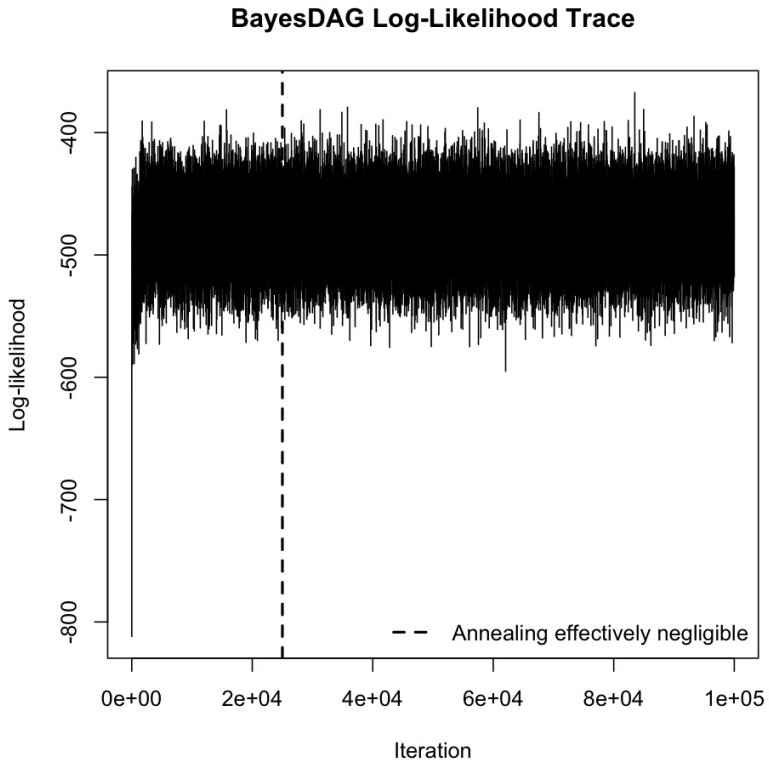


Figure 1: Representative `BayesDAG` log-likelihood trace for $p = 20$, $N = 100$, with Laplace errors. The vertical dashed line marks the point at which the annealing contribution becomes negligible.

Convergence of `BayesDCG`. Figure 2 presents the corresponding log-likelihood trajectory for `BayesDCG` under the same simulation setting. The cyclic sampler first explores the unrestricted graph space before transitioning into the admissible disjoint cycle space. The first

vertical line marks this transition, where the current state is projected into the constrained model space using the procedure described in Section C.4. The second vertical line marks the end of the post-projection annealing period.

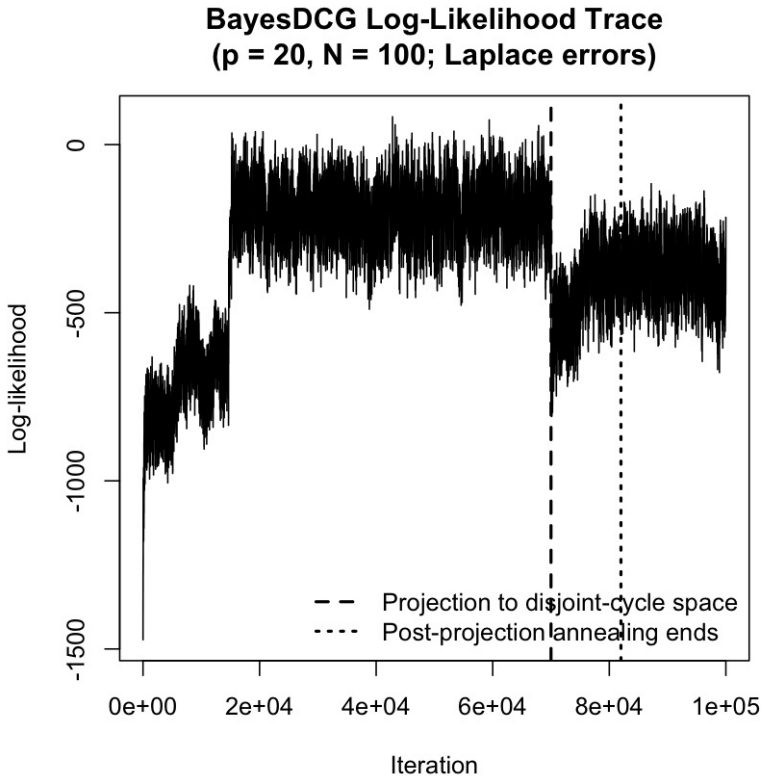
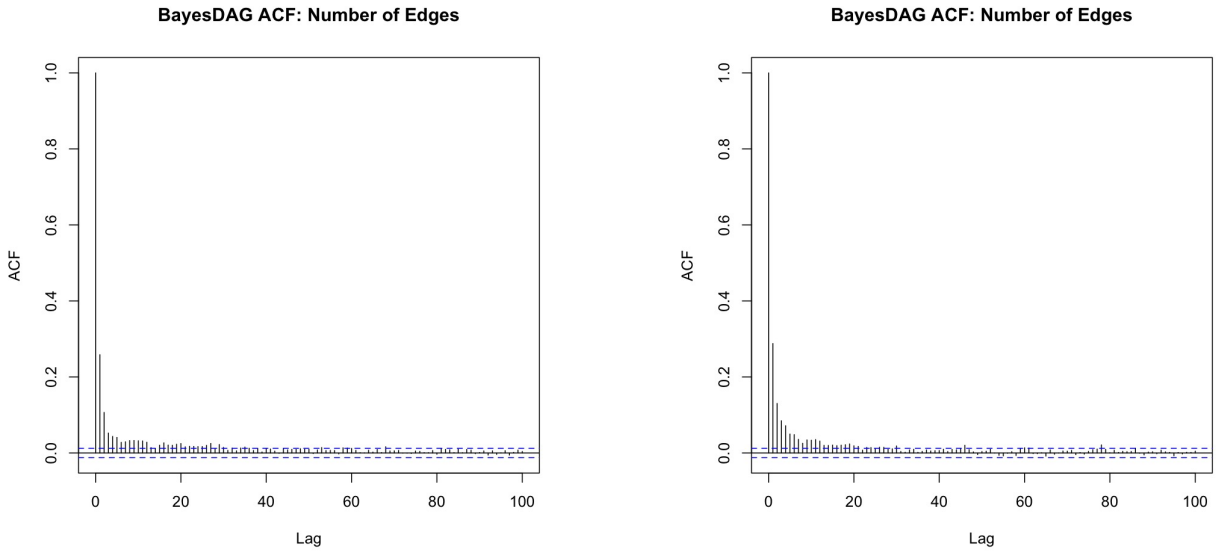


Figure 2: Representative `BayesDCG` log-likelihood trace for $p = 20$, $N = 100$, with Laplace errors. The first vertical line marks the transition into the disjoint cycle constrained space, and the second marks the end of the post-projection annealing period.

The shift in log-likelihood at the first transition reflects the projection into the constrained graph space. After the post-projection annealing period, the log-likelihood fluctuates around a relatively stable level, providing empirical evidence that `BayesDCG` has entered a stable sampling regime during the retained posterior-sampling phase.

Mixing diagnostics. Although the preceding trajectories provide empirical convergence diagnostics, their stabilization alone does not establish satisfactory MCMC mixing. We therefore examine the autocorrelation function (ACF) of the number of edges in the sampled graphs, using only draws from the retained posterior-sampling portion of each chain.

Figure 3 presents representative `BayesDAG` ACFs for $p = 20$ with $N = 100$ and $N = 200$ under Laplace errors.

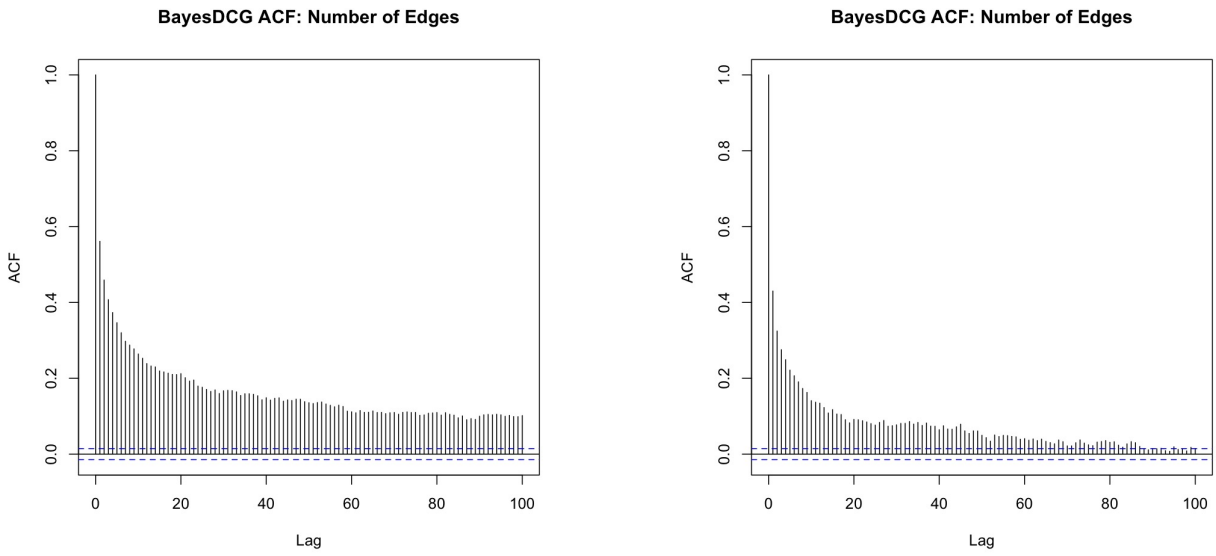


(a) $N = 100$. **(b)** $N = 200$.

Figure 3: Autocorrelation functions for the number of sampled edges from representative `BayesDAG` simulations with $p = 20$ under Laplace errors.

In both settings, autocorrelation is strongest at the first few lags and decreases rapidly toward zero. We therefore interpret these diagnostics as indicating good graph-level mixing for `BayesDAG` in the representative settings examined.

Figure 4 presents the corresponding `BayesDCG` diagnostics.



(a) $N = 100$. **(b)** $N = 200$.

Figure 4: Autocorrelation functions for the number of sampled edges from representative `BayesDCG` simulations with $p = 20$ under Laplace errors.

The `BayesDCG` chains exhibit more persistent short-range autocorrelation than `BayesDAG`, although the dependence gradually decreases toward zero as the lag increases. These results indicate slower, but still adequate, graph-level mixing during the retained posterior sampling phase.

Overall, the trace plots and edge count ACFs provide complementary empirical evidence of stable posterior exploration for both samplers. However, because the number of edges is only a low dimensional summary of the sampled graph, these diagnostics do not fully characterize mixing over the entire posterior distribution.

7 Conclusion

We introduce `cyclinbayes`, an R package that provides a unified Bayesian framework for causal discovery under linear non-Gaussian structural equation models. The package im-

plements two complementary methods: `BayesDAG` for acyclic graphs and `BayesDCG` for directed cyclic graphs with feedback. Both jointly learn graph structures, causal effect coefficients, sparsity parameters, and finite-mixture error distributions through MCMC. The Beta–Bernoulli and spike-and-slab priors provide sparsity and model complexity control, while structural restrictions are incorporated directly into posterior sampling.

Beyond estimating a single graph, `cyclinbayes` provides posterior uncertainty quantification for direct causal effects and user-specified network motifs, allowing researchers to assess the sensitivity of structural conclusions across plausible graph configurations. The proposed Bayesian framework also offers benefits for graph recovery: our simulation studies demonstrate improved false-positive control and a favorable balance between precision and recall across the considered non Gaussian error settings, particularly in smaller sample problems. We further introduce a decision-theoretic graph summarization approach that selects a representative structure by minimizing posterior expected loss under global discrepancy measures such as SHD and SID. Together with its Rcpp implementation, these features make `cyclinbayes` a practical toolbox for causal discovery that combines graph recovery, structural constraints, and posterior inference within a unified workflow.

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APPENDIX

This appendix consists of two sections: Sections A and B contain derivations of full conditional distributions under DAG and under DCG, respectively.

A Prior Properties and Multiplicity Adjustment

The hierarchical Beta-Bernoulli prior used in `BayesDAG` and `BayesDCG` provides an adaptive prior on graph sparsity and also induces a form of multiplicity adjustment. This idea is closely related to the Bayesian variable selection framework of Scott and Berger [2010], who show that placing a prior on a common inclusion probability can automatically account for multiplicity as the number of candidate variables increases. Here, we apply the same principle to edge inclusion in causal graph learning.

Multiplicity adjustment under the Beta–Bernoulli prior. Recall that, conditional on a common edge inclusion probability γ,

$$E_{ij} \mid \gamma \sim \text{Bernoulli}(\gamma), \qquad \gamma \sim \text{Beta}(a_\gamma, b_\gamma),$$

for $i \neq j$. Let

$$K = p(p-1)$$

denote the total number of candidate directed edges when self-loops are excluded, and let

$$m = \sum_{i \neq j} E_{ij}$$

denote the number of included edges in a particular adjacency matrix E. Conditional on γ,

$$p(E \mid \gamma) = \gamma^m (1-\gamma)^{K-m}.$$

Integrating over γ gives

$$\begin{aligned} p(E) &= \int_0^1 p(E \mid \gamma) p(\gamma)\, d\gamma \\ &= \frac{1}{\mathrm{B}(a_\gamma, b_\gamma)} \int_0^1 \gamma^{a_\gamma + m - 1} (1-\gamma)^{b_\gamma + K - m - 1}\, d\gamma \\ &= \frac{\mathrm{B}(a_\gamma + m, b_\gamma + K - m)}{\mathrm{B}(a_\gamma, b_\gamma)}. \end{aligned}$$

Thus, after integrating out γ, the marginal prior on E depends on the total graph size m. Because all candidate edges share the same unknown inclusion probability, the prior adapts to the overall sparsity of the graph rather than assigning each edge a fixed inclusion probability independently. This induces a form of automatic multiplicity adjustment analogous to that described by Scott and Berger [2010] for Bayesian variable selection.

The effect is particularly clear in the illustrative case $a_\gamma = b_\gamma = 1$, for which

$$p(E) = \frac{1}{K+1} \frac{1}{\binom{K}{m}}.$$

Since there are $\binom{K}{m}$ unrestricted adjacency matrices containing exactly m directed edges,

$$P(|E| = m) = \binom{K}{m} p(E) = \frac{1}{K+1}.$$

Hence, graph sizes with more possible edge configurations do not receive greater total prior probability simply because there are more graphs of that size. The factor $1/\binom{K}{m}$ offsets this combinatorial multiplicity.

Bayesian Occam's razor. The multiplicity adjustment induced by the Beta–Bernoulli prior is distinct from the Bayesian Occam's-razor effect arising through marginal likelihoods. For two candidate graphs E_0 and E_1, the posterior odds can be written as

$$\frac{p(E_1 \mid \mathbf{Y})}{p(E_0 \mid \mathbf{Y})} = \underbrace{\frac{p(\mathbf{Y} \mid E_1)}{p(\mathbf{Y} \mid E_0)}}_{\text{Bayes factor}} \underbrace{\frac{p(E_1)}{p(E_0)}}_{\text{prior odds}},$$

where

$$p(\mathbf{Y} \mid E) = \int p(\mathbf{Y} \mid B, E) p(B \mid E)\, dB.$$

A graph with additional edges has greater flexibility and may achieve a larger likelihood at its best fitting parameter values. However, the marginal likelihood averages over the additional parameter space. An extra edge is therefore favored only when its improvement in fit is sufficient to overcome the additional model complexity. This is the Bayesian Occam's-razor effect.

In `BayesDAG`, this mechanism arises directly because the causal effect coefficients are analytically marginalized during graph updates. Candidate graphs are therefore compared using integrated evidence rather than the likelihood at a single value of B, see Section E.

The two mechanisms can be summarized as

$$\underbrace{p(E)}_{\text{Beta–Bernoulli prior}} \quad \longrightarrow \quad \text{multiplicity adjustment and graph-size regularization,}$$

and

$$\underbrace{p(\mathbf{Y} \mid E)}_{\text{marginal likelihood}} \quad \longrightarrow \quad \text{Occam's-razor penalty for unnecessary complexity.}$$

Together with the spike-and-slab prior on B, these mechanisms favor parsimonious graph structures while allowing additional edges when they are sufficiently supported by the data.

B Derivation of Full Conditional Distributions Under DAG

In this section, we derive the full conditional distribution of each parameter in the collapsed Gibbs sampler for DAG.

B.1 Full Conditional distribution of γ

$$
\begin{aligned}
[\gamma|-] &\propto [\gamma][E_{21}|\gamma]...[E_{pp-1}|\gamma] \\
&= \gamma^{a_\gamma+\sum_{j=2}^{p}\sum_{l\in pa(j)}E_{jl}-1}(1-\gamma)^{b_\gamma+\sum_{j=2}^{p}\sum_{l\in pa(j)}(1-E_{jl})-1} \\
&\sim Beta\left(a_\gamma+\sum_{j=2}^{p}\sum_{l\in pa(j)}E_{jl}, b_\gamma+\sum_{j=2}^{p}\sum_{l\in pa(j)}(1-E_{jl})\right)
\end{aligned} \tag{γ}
$$

B.2 Full Conditional distribution of γ_1

$$
\begin{aligned}
[\gamma_1|-] &\propto [\gamma_1][B_{21}|E_{21},\gamma_1]...[B_{pp-1}|E_{pp-1},\gamma_1] \\
&= \frac{b_\gamma^{a_\gamma}}{\Gamma(a_\gamma)}(\gamma_1)^{-(a_\gamma+1)}\exp\left(\frac{-b_\gamma}{\gamma_1}\right)\times\prod_{i=2,...,p,\, j\in pa(i)}\{I(B_{ij}=0)\}^{1-E_{ij}}\{N(0,\gamma_1)\}^{E_{ij}} \\
&= \frac{b_\gamma^{a_\gamma}}{\Gamma(a_\gamma)}(\gamma_1)^{-(a_\gamma+1)}\exp\left(\frac{-b_\gamma}{\gamma_1}\right) \\
&\times\prod_{i=2,...,p,\, j\in pa(i)}\{I(B_{ij}=0)\}^{1-E_{ij}}\{(\frac{1}{\sqrt{2\pi\gamma_1}})^{E_{ij}}\exp\left(-\frac{E_{ij}}{2\gamma_1}(B_{ij})^2\right)\}
\end{aligned} \tag{γ_1}
$$

$$
\begin{aligned}
&\propto (\gamma_1)^{-(a_\gamma+1)}\exp\left(\frac{-b_\gamma}{\gamma_1}\right)\prod_{i=2,...,p,\, j\in pa(i)}\exp\left(-\frac{E_{ij}}{2\gamma_1}B_{ij}^2\right)(\frac{1}{\sqrt{2\pi\gamma_1}})^{E_{il}} \\
&\times(\gamma_1)^{-(a_\gamma+\frac{\sum_{i=2}^{p}\sum_{j\in pa(i)}E_{ij}}{2}+1)}\exp\left(-\frac{b_\gamma+\frac{1}{2}\sum_{i=2}^{p}\sum_{j\in pa(i)}(E_{ij})(B_{ij})^2}{\gamma_1}\right) \\
&\sim IG\left(a_\gamma+\frac{\sum_{i=2}^{p}\sum_{j\in pa(i)}E_{ij}}{2}, b_\gamma+\frac{1}{2}\sum_{i=2}^{p}\sum_{j\in pa(i)}(E_{ij})(B_{ij})^2\right)
\end{aligned}
$$

B.3 Full conditional distribution of $Z_i^{(q)}$

$$
\begin{aligned}
&[Z_i^{(q)}|\,\pi_{i1},...,\pi_{iM}][\epsilon_1^{(1)},\cdots,\epsilon_p^{(1)},\cdots,\epsilon_1^{(q)},\cdots,\epsilon_p^{(N)}|all\,\pi, all\,\mu, all\,\tau] \\
&\propto \pi_{i1}^{I(Z_i^{(q)}=1)}...\pi_{iG}^{I(Z_i^{(q)}=M)}\prod_{k=1}^{M}\{N(\epsilon_i^{(q)},\mu_{ik},\tau_{ik})\}^{I(Z_i^{(q)}=k)} \\
&\{\pi_{i1}\,N(\epsilon_i^{(q)},\mu_{i1},\tau_{i1})\}^{I(Z_i^{(q)}=1)}\cdots\{\pi_{jM}N(\epsilon_i^{(q)},\mu_{iM},\tau_{iM})\}^{I(Z_i^{(q)}=M)} \\
&\sim Categorical\left(\pi_{i1}N(\epsilon_i^{(q)},\mu_{i1},\tau_{i1}),\cdots,\pi_{iM}N(\epsilon_i^{(q)},\mu_{iM},\tau_{iM})\right)
\end{aligned} \tag{$Z_i^{(q)}$}
$$

B.4 Full conditional distribution of π_{ik}

$$
\begin{aligned}
&[\vec{\pi_i}][\mathbf{Z}_i|\vec{\pi_i}] \propto \prod_{q=1}^{N}\prod_{k=1}^{M}(\pi_{ik})^{\alpha-1}\{(\pi_{i1})^{I(Z_i^{(q)}=1)}...(\pi_{iM})^{I(Z_j^{(q)}=M)}\} \\
&\pi_{i1},...,\pi_{ik},...,\pi_{iM} \sim Dir(\alpha+\sum_{q=1}^{N} I(Z_i^{(q)}=1),...,\alpha+\sum_{q=1}^{N} I(Z_i^{(q)}=M))
\end{aligned}
\tag{π_{ik}}
$$

B.5 Full conditional distribution of μ_{ik}

$$
\begin{aligned}
&[\mu_{ik}][\epsilon_1^{(1)}, \cdots, \epsilon_p^{(1)}, \cdots, \epsilon_1^{(N)}, \cdots, \epsilon_p^{(N)} \mid all\,\pi, all\,\mu, all\,\tau] \\
&\quad \propto \frac{1}{\sqrt{2\pi b_\mu}} \exp\left(-\frac{1}{2b_\mu}(\mu_{ik} - a_\mu)^2\right) \prod_{q=1}^{N} \prod_{i'=1}^{p} \prod_{k'=1}^{G} N(\epsilon_i^{(q)}, \mu_{i'k'}, \tau_{jk})^{I(Z_i^{(q)}=k')} \\
&\quad = \frac{1}{\sqrt{2\pi b_\mu}} \exp\left(-\frac{1}{2b_\mu}(\mu_{ik} - a_\mu)^2\right) \prod_{q=1}^{N} \{N(\epsilon_i^{(q)}, \mu_{ik}, \tau_{ik})\}^{I(Z_i^{(q)}=k)} \\
&\quad \propto \exp\left(-\frac{1}{2b_\mu}((\mu_{ik})^2 - 2a_\mu\mu_{ik})\right) \prod_{q=1}^{N} \left\{\left(\frac{1}{\sqrt{2\pi\tau_{ik}}}\right) \exp\left(-\frac{1}{2\tau_{ik}}(\epsilon_i^{(q)} - \mu_{ik})^2\right)\right\}^{I(Z_i^{(q)}=k)} \\
&\quad \propto \exp\left(-\frac{1}{2b_\mu}\{(\mu_{ik})^2 - 2a_\mu\mu_{ik}\}\right) \left(\frac{1}{\sqrt{2\pi\tau_{ik}}}\right)^{\sum_{q=1}^{N} I(Z_i^{(q)}=k)} \\
&\qquad \cdot \exp\left(-\sum_{q=1}^{N} \frac{I(Z_i^{(q)}=k)}{2\tau_{ik}}\{(\mu_{ik})^2 - 2\mu_{ik}\epsilon_i^{(q)}\}\right) \\
&\quad \propto \exp\left(\left\{-\frac{1}{2b_\mu} - \frac{\sum_{q=1}^{N} I(Z_i^{(q)}=k)}{2\tau_{ik}}\right\}(\mu_{ik})^2 - 2\left\{-\frac{1}{2b_\mu}a_\mu\mu_{ik} - \sum_{q=1}^{N}\frac{I(Z_i^{(q)}=k)}{2\tau_{ik}}\mu_{ik}\epsilon_i^{(q)}\right\}\right) \\
&\quad = \exp\left(-\frac{1}{2}\left\{\frac{1}{b_\mu} + \frac{\sum_{q=1}^{N} I(Z_i^{(q)}=k)}{\tau_{ik}}\right\}(\mu_{ik})^2 + \left\{\frac{a_\mu}{b_\mu} + \sum_{q=1}^{N}\frac{I(Z_i^{(q)}=k)}{\tau_{ik}}\epsilon_i^{(q)}\right\}\mu_{ik}\right) \\
&\quad = \exp\left(-\frac{1}{2}\left\{\frac{1}{b_\mu} + \frac{\sum_{q=1}^{N} I(Z_i^{(q)}=k)}{\tau_{ik}}\right\}\left\{(\mu_{ik})^2 - 2\frac{\frac{a_\mu}{b_\mu} + \sum_{q=1}^{N}\frac{I(Z_i^{(q)}=k)}{\tau_{ik}}\epsilon_i^{(q)}}{\frac{1}{b_\mu} + \frac{\sum_{q=1}^{N} I(Z_i^{(q)}=k)}{\tau_{ik}}}\mu_{ik}\right\}\right) \\
&\quad \propto \exp\left(-\frac{1}{2}\left\{\frac{1}{b_\mu} + \frac{\sum_{q=1}^{N} I(Z_i^{(q)}=k)}{\tau_{ik}}\right\}\left\{\mu_{ik} - \frac{\frac{a_\mu}{b_\mu} + \sum_{q=1}^{N}\frac{I(Z_i^{(q)}=k)}{\tau_{ik}}\epsilon_i^{(q)}}{\frac{1}{b_\mu} + \frac{\sum_{q=1}^{N} I(Z_i^{(q)}=k)}{\tau_{ik}}}\right\}^2\right) \\
&\quad \sim N\left(\frac{\frac{a_\mu}{b_\mu} + \sum_{q=1}^{N}\frac{I(Z_i^{(q)}=k)}{\tau_{ik}}\epsilon_i^{(q)}}{\frac{1}{b_\mu} + \frac{\sum_{q=1}^{N} I(Z_i^{(q)}=k)}{\tau_{ik}}},\ \frac{1}{\frac{1}{b_\mu} + \frac{\sum_{q=1}^{N} I(Z_i^{(q)}=k)}{\tau_{ik}}}\right)
\end{aligned}
$$

(μ_{ik})

B.6 Full conditional distribution of τ_{ik}

$$
\begin{aligned}
&[\tau_{ik}][\epsilon_1^{(1)}, \cdots, \epsilon_p^{(1)}, \cdots, \epsilon_1^{(N)}, \cdots, \epsilon_p^{(N)} \mid all\ \pi, all\ \mu, all\ \tau] \\
&\quad \propto \tau_{ik}^{-(a_\tau+1)} \exp\left(-\frac{b_\tau}{\tau_{ik}}\right) \prod_{q=1}^{N} \prod_{i'=1}^{p} \prod_{k'=1}^{M} N(\epsilon_i^{(q)}, \mu_{i'k'}, \tau_{i'k'})^{I(Z_j^{(q)}=k')} \\
&\quad \propto \tau_{ik}^{-(a_\tau+1)} \exp\left(-\frac{b_\tau}{\tau_{ik}}\right) \prod_{q=1}^{N} N(\epsilon_i^{(q)}, \mu_{ik}, \tau_{ik})^{I(Z_i^{(q)}=k)} \\
&\quad \propto \tau_{ik}^{-(a_\tau+1)} \exp\left(-\frac{b_\tau}{\tau_{ik}}\right) \prod_{q=1}^{N} \left\{ \left(\frac{1}{\sqrt{2\pi\tau_{ik}}}\right) \exp\left(-\frac{1}{2\tau_{ik}}(\epsilon_i^{(q)} - \mu_{ik})^2\right) \right\}^{I(Z_i^{(q)}=k)} \\
&\quad \propto \tau_{ik}^{-\left(a_\tau + \frac{\sum_{q=1}^{N} I(Z_i^{(q)}=k)}{2} + 1\right)} \exp\left(-\frac{b_\tau}{\tau_{ik}} - \sum_{q=1}^{N} \frac{I(Z_i^{(q)}=k)}{2\tau_{ik}} (\epsilon_i^{(q)} - \mu_{ik})^2 \right) \\
&\quad \propto \tau_{ik}^{-\left(a_\tau + \frac{\sum_{q=1}^{N} I(Z_i^{(q)}=k)}{2} + 1\right)} \exp\left(-\frac{1}{\tau_{ik}} \left[b_\tau + \sum_{q=1}^{N} \frac{I(Z_i^{(q)}=k)}{2} (\epsilon_i^{(q)} - \mu_{ik})^2 \right] \right) \\
&[\tau_{ik} \mid -] \propto IG\left(a_\tau + \frac{\sum_{q=1}^{N} I(Z_i^{(q)}=k)}{2},\ b_\tau + \sum_{q=1}^{N} \frac{I(Z_i^{(q)}=k)}{2} (\epsilon_i^{(q)} - \mu_{ik})^2 \right)
\end{aligned}
\tag{τ_{ik}}
$$

B.7 Full conditional Distribution of B_{ij}

$$
\begin{aligned}
&[B_{ij} \mid E_{ij}, \gamma_1][y_i^{(1)}, \ldots, y_i^{(N)} \mid B_{i1}, \ldots, B_{ii-1}, B_{ii+1}, \ldots, B_{ip}, \epsilon_i, Z_i^{(q)}, \\
&\qquad\qquad y_1^{(1)}, \ldots, y_{i-1}^{(1)}, y_{i+1}^{(1)}, \ldots, y_p^{(1)}, \ldots, y_1^{(N)}, \ldots, y_{i-1}^{(N)}, y_{i+1}^{(N)}, \ldots, y_p^{(N)}] \\
&\propto [E_{ij}(N(0,\gamma_1)) + (1-E_{ij})(I(B_{ij}=0))] \left[\prod_{q=1}^{N} \prod_{k=1}^{M} \{N(\mu_{ik} + \sum_{j' \in pa(i)} B_{ij'} Y_{j'}^{(q)}, \tau_{ik})\}^{I(Z_i^{(q)}=k)} \right] \\
&\propto [E_{ij}(\exp(-\frac{E_{ij}}{2\gamma_1} B_{ij}^2)) + (1-E_{ij})(I(B_{ij}=0))] \\
&[\exp(-\sum_{q=1}^{N}\sum_{k=1}^{M} \frac{I(Z_i^{(q)}=k)}{2\tau_{ik}} \{y_i^{(q)} - (\mu_{ik} + \sum_{j' \in pa(i)} B_{ij'} Y_{j'}^{(q)})\}^2)] \\
&\propto [E_{ij}(\exp(-\frac{E_{ij}}{2\gamma_1} B_{ij}^2)) + (1-E_{ij})(I(B_{ij}=0))] \exp\Big(-\sum_{q=1}^{N}\sum_{k=1}^{M} \frac{I(Z_i^{(q)}=k)}{2\tau_{ik}} \{(y_i^{(q)})^2 \\
&\quad - 2y_i^{(q)}(\mu_{ik} + \sum_{j' \in pa(i)} B_{ij'} Y_{j'}^{(q)}) + (\mu_{ik} + \sum_{j' \in pa(i)} B_{ij'} Y_{j'}^{(q)})^2\}\Big) \\
&\propto [E_{ij}(\exp(-\frac{E_{ij}}{2\gamma_1} B_{ij}^2)) + (1-E_{ij})(I(B_{ij}=0))] \exp\Big(-\sum_{q=1}^{N}\sum_{k=1}^{M} \frac{I(Z_i^{(q)}=k)}{2\tau_{ik}} \{2\mu_{ik} \sum_{j' \in pa(i)} B_{ij'} Y_{j'}^{(q)} \\
&\quad + (\sum_{j' \in pa(i)} B_{ij'} Y_{j'}^{(q)})^2 - 2y_i^{(q)} \sum_{j' \in pa(i)} B_{ij'} Y_{j'}^{(q)})\}\Big) \\
&\propto [E_{ij}(\exp(-\frac{E_{ij}}{2\gamma_1} B_{ij}^2)) + (1-E_{ij})(I(B_{ij}=0))] \exp\Big(-\sum_{q=1}^{N}\sum_{k=1}^{M} \frac{I(Z_i^{(q)}=k)}{2\tau_{ik}} \{(B_{ij} Y_j^{(q)})^2 \\
&\quad + 2\mu_{ik} B_{ij} y_j^{(q)} + 2\sum_{j' \neq j} B_{ij} B_{ij'} Y_j^{(q)} Y_{j'}^{(q)} - 2B_{ij} y_i^{(q)} y_j^{(q)}\}\Big) \\
&= [E_{ij}(\exp(-\frac{E_{ij}}{2\gamma_1} B_{ij}^2)) + (1-E_{ij})(I(B_{ij}=0))] \exp\Big(\{-\sum_{q=1}^{N}\sum_{k=1}^{M} (y_j^{(q)})^2 \frac{I(Z_i^{(q)}=k)}{2\tau_{ik}}\} B_{ij}^2 \\
&\quad + 2B_{ij} \sum_{q=1}^{N}\sum_{k=1}^{M} y_j^{(q)} \frac{I(Z_i^{(q)}=k)}{2\tau_{ik}} (-\mu_{ik} - \sum_{j' \neq j} B_{ij'} Y_{j'}^{(q)} + y_i^{(q)})\Big).
\end{aligned}
\tag{B_{ij}}
$$

Note: since we need to obtain the pdf for B_{ij}, $\sum_{l \in pa(i), l \neq j} B_{il}$ can be considered as a constant. This means that $\exp(-\{(-2y_i^{(q)} \sum_{j' \in pa(i)} B_{ij'} Y_{j'}^{(q)})\}) = \exp(-\{(-2y_i^{(q)} B_{ij} Y_j^{(q)})\})$.

Therefore, if $E_{ij} = 0$, then $B_{ij} = 0$;

Else if $B_{ij} \neq 0$,

$$[B_{ij} \mid -] \propto N\left(\frac{\sum_{q=1}^{N}\sum_{k=1}^{M} y_j^{(q)} \frac{I(Z_i^{(q)}=k)}{\tau_{ik}}(y_i^{(q)} - (\mu_{ik} + \sum_{j' \neq j} B_{ij'} Y_{j'}^{(q)}))}{\frac{1}{\gamma_1} + \sum_{q=1}^{N}\sum_{k=1}^{M}(y_j^{(q)})^2 \frac{I(Z_i^{(q)}=k)}{\tau_{ik}}}, \frac{1}{\frac{1}{\gamma_1} + \sum_{q=1}^{N}\sum_{k=1}^{M}(y_j^{(q)})^2 \frac{I(Z_i^{(q)}=k)}{\tau_{ik}}}\right).$$

B.8 E Full Conditional

B.8.1 Marginal conditional Distribution of E by integrating out B

$$\begin{aligned}
[E_{21}, \ldots, E_{ij}, \ldots, E_{pp-1} \mid -] &\propto [Y_1^{(1)}, \cdots, Y_p^{(1)}, \cdots, Y_1^{(N)}, \cdots, Y_p^{(N)} \mid E_{21}, \cdots, E_{ij}, \cdots, E_{pp-1}] \times [E \mid \gamma] \\
&= \int\int \ldots \int \{[Y_1^{(1)}, \cdots, Y_p^{(1)}, \cdots, Y_1^{(N)}, \cdots, Y_p^{(N)} \mid B_{21}, ..., B_{pp-1}] \\
&\quad \times [B_{21} \mid E_{21}, \gamma_1] \ldots [B_{pp-1} \mid E_{pp-1}, \gamma_1]\}\, dB_{21} \ldots dB_{pp-1} \times [E_{21}, \ldots, E_{ij}, \ldots, E_{pp-1} \mid \gamma].
\end{aligned} \tag{E}$$

Let $Pa(i) = \cup_{l'=0}^{p-1}\{(j_1, \ldots, j_{l'}) : 1 \leq j_1 < \ldots < j_{l'} \leq p,\, j_m \neq i, \text{ for } m = 1, \cdots, l'\}$. For a given $(j_1, \ldots, j_l) \in Pa(i)$, we can re-express our model as follows:

$$\begin{aligned}
\vec{Y_i} = \begin{bmatrix} Y_i^{(1)} \\ Y_i^{(2)} \\ \vdots \\ Y_i^{(N)} \end{bmatrix} &= \begin{bmatrix} Y_{j_1}^{(1)} & Y_{j_2}^{(1)} & \ldots & Y_{j_l}^{(1)} \\ Y_{j_1}^{(2)} & Y_{j_2}^{(2)} & \ldots & Y_{j_l}^{(2)} \\ \vdots & \vdots & \ddots & \vdots \\ Y_{j_1}^{(N)} & Y_{j_2}^{(N)} & \ldots & Y_{j_l}^{(N)} \end{bmatrix} \begin{bmatrix} B_{ij_1} \\ B_{ij_2} \\ \vdots \\ B_{ij_l} \end{bmatrix} + \begin{bmatrix} \epsilon_i^{(1)} \\ \epsilon_i^{(2)} \\ \vdots \\ \epsilon_i^{(N)} \end{bmatrix} \\
&= \begin{bmatrix} \vec{Y_{j_1}} & \vec{Y_{j_2}} & \ldots & \vec{Y_{j_l}} \end{bmatrix} \vec{B_i} + \vec{\epsilon_i} \\
&\sim D_i \vec{B_i} + MVN(\vec{\mu_i}, \vec{\tau_i} I) \quad \text{for } i = 1, \ldots, p \text{ and } D_i = (Y_{j_1}, \ldots, Y_{j_l}).
\end{aligned}$$

$$\begin{bmatrix} \epsilon_i^{(1)} \\ \epsilon_i^{(2)} \\ \vdots \\ \epsilon_i^{(N)} \end{bmatrix} \sim MVN\left(\begin{bmatrix} \mu_{ik_{(1)}} \\ \mu_{ik_{(2)}} \\ \vdots \\ \mu_{ik_{(N)}} \end{bmatrix}, \begin{bmatrix} \tau_{ik_{(1)}} \\ \tau_{ik_{(2)}} \\ \vdots \\ \tau_{ik_{(N)}} \end{bmatrix} I\right), \quad \text{where } k_{(q)} = 1, \ldots, G \text{ satisfies } Z_{ik}^{(q)} = 1 \text{ if } k = k_{(q)}.$$

The multivariate priors for the vector B are

$$\left[\begin{bmatrix} B_{ij_1} \\ B_{ij_2} \\ \vdots \\ B_{ij_l} \end{bmatrix} \middle| \middle| \begin{bmatrix} E_{ij_1} \\ E_{ij_2} \\ \vdots \\ E_{ij_l} \end{bmatrix}, \gamma_1 \right] = [\vec{B}_i \mid \vec{E}_i, \gamma_1] \sim MVN(0, \gamma_1 I). \tag{1}$$

Then the marginal conditional distribution of the ith row of E by integrated the ith row of B is

$$\begin{aligned}
&[E_{i1}, \ldots, E_{ii-1}, E_{ii+1}, \ldots E_{ip} \mid -] \\
&\propto [Y_1^{(1)}, \cdots, Y_p^{(1)}, \cdots, Y_1^{(N)}, \cdots, Y_p^{(N)} \mid E_{i1}, \ldots, E_{ii-1}, E_{ii+1}, \ldots E_{ip}] \times [E_{i1}, \ldots, E_{ii-1}, E_{ii+1}, \ldots E_{ip} \mid \gamma] \\
&= \sum_{l=0}^{p-1} \sum_{(j_1,\ldots,j_l) \in Pa(i)} [\prod_{j' \neq j_1,\ldots,j_l} (1 - E_{ij'})][\prod_{m=1}^{l} E_{ij_m}] \\
&\quad \times \int \int \ldots \int \{[Y_{j_1}^{(1)}, \cdots, Y_{j_l}^{(1)}, \cdots, Y_{j_1}^{(q)}, \cdots, Y_{j_l}^{(q)}, \cdots, Y_{j_1}^{(N)}, \cdots, Y_{j_l}^{(N)} \mid B_{ij_1}, \ldots, B_{ij_l}] \\
&\quad \times [B_{ij_1} \mid E_{ij_1}, \gamma_1] \ldots [B_{ij_l} \mid E_{ij_l}, \gamma_1]\} \, dB_{ij_1} \ldots dB_{ij_l} \times [E_{i1}, \ldots, E_{ii-1}, E_{ii+1}, \ldots E_{ip} \mid \gamma] \\
&= \sum_{l=0}^{p-1} \sum_{(j_1,\ldots,j_l) \in Pa(i)} [\prod_{j' \neq j_1,\ldots,j_l} (1 - E_{ij'})][\prod_{m=1}^{l} E_{ij_m}] \\
&\quad \times \int \int \ldots \int \{[Y_1^{(1)}, \cdots, Y_p^{(1)}, \cdots, Y_1^{(N)}, \cdots, Y_p^{(N)} \mid \vec{B}_i] \\
&\quad \times [\vec{B}_i \mid \vec{E}_i, \gamma_1] d\vec{B}_i] \times [E_{i1}, \ldots, E_{ii-1}, E_{ii+1}, \ldots E_{ip} \mid \gamma] \\
&= \sum_{l=0}^{p-1} \sum_{(j_1,\ldots,j_l) \in Pa(i)} [\prod_{j' \neq j_1,\ldots,j_l} (1 - E_{ij'})][\prod_{m=1}^{l} E_{ij_m}] \\
&\quad \times \int \int \ldots \int \{[Y_1^{(1)}, \cdots, Y_p^{(1)}, \cdots, Y_1^{(N)}, \cdots, Y_p^{(N)} \mid \vec{B}_i] \\
&\quad \times [\vec{B}_i \mid \vec{E}_i, \gamma_1][\vec{E}_i \mid \gamma] d\vec{B}_i] \\
&= \sum_{l=0}^{p-1} \sum_{(j_1,\ldots,j_l) \in Pa(i)} [\prod_{j' \neq j_1,\ldots,j_l} (1 - E_{ij'})][\prod_{m=1}^{l} E_{ij_m}] \times \mathbf{I}^i_{j_1,\ldots,j_l},
\end{aligned}$$

where $(j_1, \ldots, j_l) \in Pa(i) = \cup_{l'=0}^{p-1} \{(j_1, \ldots, j_{l'}) : 1 \leq j_1 < \ldots < j_{l'} \leq p, \, j_m \neq i, \text{ for } m = 1, \cdots, l'\}$. Here, the detailed calculation of $\mathbf{I}^{\mathbf{i}}_{\mathbf{j_1},\ldots,\mathbf{j_l}}$ is described in Section B.9.

Let $\mathcal{I}$ denote the set of all possible configurations of $(E_{21}, \ldots, E_{p,p-1})$, such that

$$\mathcal{I} = \{(E_{21}, \ldots, E_{p,p-1}) \mid E_{ij} \in \{0,1\},\ 1 \leq i,j \leq p,\ i \neq j\}\,.$$

The total number of elements in $\mathcal{I}$ is $2^{p(p-1)}$. The probability that $(E_{21}, \ldots, E_{p,p-1} \mid -) = I_e$, for some $I_e \in \mathcal{I}$, is proportional to $\tilde{P}_{I_e}$, where $\tilde{P}_{I_e}$ is defined in equation (E).

Define

$$P_{I_e} = \frac{\tilde{P}_{I_e}}{\sum_{I_e \in \mathcal{I}} \tilde{P}_{I_e}}.$$

Then, $[(E_{21}, \ldots, E_{pp-1}) = I_e \mid -] \sim \text{Categorical}(p_{I_e} : I_e \in \mathcal{I})$ which has a closed form.

However, direct sampling via Gibbs sampling using this closed form is infeasible due to the size of $\mathcal{I}$, which grows exponentially as $p \to \infty$. Hence, instead of using a Gibbs sampler, we use MH for sampling E.

B.9 Calculation of $\mathbf{I}^{\mathbf{i}}_{\mathbf{j_1},\ldots,\mathbf{j_l}}$, r, and $\tilde{r}$

$$\begin{aligned}
\mathbf{I}^{\mathbf{i}}_{\mathbf{j_1},\ldots,\mathbf{j_l}} &= \int [\vec{Y}_i | \vec{B}_i, \vec{\mu}_i, \vec{\tau}_i][\vec{B}_i | \vec{E}_i, \gamma_1] d\vec{B}_i \\
&= \int (2\pi)^{-N/2} (\det |\vec{\tau}_i I|)^{-1/2} \times \exp\Big(-\frac{1}{2}(\vec{Y}_i - (\vec{\mu}_i + D_i\vec{B}_i))^T (\vec{\tau}_i I)^{-1} (\vec{Y}_i - (\vec{\mu}_i + D_i\vec{B}_i))\Big) \\
&\quad \times (2\pi)^{-l/2} (\det |\vec{\gamma_1} I|)^{-1/2} \exp\Big(-\frac{1}{2}\vec{B}_i^T (\gamma_1 I)^{-1} \vec{B}_i\Big) d\vec{B}_i \\
&= (2\pi)^{-(l+N)/2} (\det |\vec{\tau}_i I|)^{-1/2} (\det |\vec{\gamma_1} I|)^{-1/2} \\
&\quad \times \int \exp\Big(\underbrace{-\frac{1}{2}(\vec{Y}_i - (\vec{\mu}_i + D_i\vec{B}_i)^T (\vec{\tau}_i I)^{-1} (\vec{Y}_i - (\vec{\mu}_i + D_i\vec{B}_i) - \frac{1}{2}\vec{B}_i^T (\gamma_1 I)^{-1} \vec{B}_i}_{\circledast}\Big) d\vec{B}_i
\end{aligned}$$

We first set $V = (\gamma_1 I_{l\times l})^{-1} + D_i^T (\vec{\tau}_i I_{N\times N})^{-1} D_i$ for $\vec{\tau}_i I_{N\times N} = diag(\tau_{ik_{(1)}}, \ldots, \tau_{ik_{(N)}})$ and

calculate

$$
\begin{aligned}
\circledast &= -\frac{1}{2}(\vec{Y}_i - (\vec{\mu}_i + D_i\vec{B}_i))^T(\vec{\tau}_i I)^{-1}(\vec{Y} - (\vec{\mu}_i + D_i\vec{B}_i)) - \frac{1}{2}\vec{B}_i^T(\gamma_1 I)^{-1}\vec{B}_i \\
&= -\frac{1}{2}((\vec{Y}_i - \vec{\mu}_i)^T(\vec{\tau}_i I)^{-1}(\vec{Y}_i - \vec{\mu}_i)) - \frac{1}{2}(\vec{B}_i^T((\gamma_1 I)^{-1} + D_i^T(\vec{\tau}_i I)^{-1}D_i)\vec{B}_i \\
&- 2\vec{B}_i^T D_i^T(\vec{\tau}_i I)^{-1}(\vec{Y}_i - \vec{\mu}_i)) \\
&= -\frac{1}{2}((\vec{Y}_i - \vec{\mu}_i)^T(\vec{\tau}_i I)^{-1}(\vec{Y}_i - \vec{\mu}_i)) - \frac{1}{2}(\vec{B}_i^T V\vec{B}_i - 2\vec{B}_i^T VV^{-1}D_i^T(\vec{\tau}_i I)^{-1}(\vec{Y}_i - \vec{\mu}_i)) \\
&= -\frac{1}{2}((\vec{Y}_i - \vec{\mu}_i)^T(\vec{\tau}_i I)^{-1}(\vec{Y}_i - \vec{\mu}_i)) - \frac{1}{2}(\vec{B}_i^T V\vec{B}_i - 2\vec{B}_i^T VV^{-1}D_i^T(\vec{\tau}_i I)^{-1}(\vec{Y}_i - \vec{\mu}_i) \\
&+ \frac{1}{2}(V^{-1}D_i^T(\vec{\tau}_i I)^{-1}(\vec{Y}_i - \vec{\mu}_i))^T V(V^{-1}D_i^T(\vec{\tau}_i I)^{-1}(\vec{Y}_i - \vec{\mu}_i)) \\
&- \frac{1}{2}(V^{-1}D_i^T(\vec{\tau}_i I)^{-1}(\vec{Y}_i - \vec{\mu}_i))^T V(V^{-1}D_i^T(\vec{\tau}_i I)^{-1}(\vec{Y}_i - \vec{\mu}_i))) \\
&= -\frac{1}{2}((\vec{Y}_i - \vec{\mu}_i)^T(\vec{\tau}_i I)^{-1}(\vec{Y}_i - \vec{\mu}_i)) \\
&\quad - \frac{1}{2}((\vec{B}_i - V^{-1}D_i^T(\vec{\tau}_i I)^{-1}(\vec{Y}_i - \vec{\mu}_i))^T V(\vec{B}_i - V^{-1}D_i^T(\vec{\tau}_i I)^{-1}(\vec{Y}_i - \vec{\mu}_i))) \\
&\quad + \frac{1}{2}(V^{-1}D_i^T(\vec{\tau}_i I)^{-1}(\vec{Y}_i - \vec{\mu}_i))^T V(V^{-1}D_i^T(\vec{\tau}_i I)^{-1}(\vec{Y}_i - \vec{\mu}_i)).
\end{aligned}
$$

Now, when we do the integral of the $\vec{B}_i$, we have

$$
\begin{aligned}
\mathbf{I}^i_{j_1,\ldots,j_l} &= (2\pi)^{-(l+N)/2}(\det|\vec{\tau}_i I|)^{-1/2}(\det|\vec{\gamma_1} I|)^{-1/2}\exp\Big(-\frac{1}{2}((\vec{Y}_i - \vec{\mu}_i)^T(\vec{\tau}_i I)^{-1}(\vec{Y}_i - \vec{\mu}_i) \\
&+ \frac{1}{2}(V^{-1}D_i^T(\tau_i I)^{-1}(\vec{Y}_i - \vec{\mu}_i))^T V(V^{-1}D_i^T(\vec{\tau}_i I)^{-1}(\vec{Y}_i - \vec{\mu}_i))\Big) \\
&\int \exp\Big(-\frac{1}{2}((\vec{B}_i - V^{-1}D_i^T(\vec{\tau}_i I)^{-1}(\vec{Y}_i - \vec{\mu}_i))^T V(\vec{B}_i - V^{-1}D_i^T(\vec{\tau}_i I)^{-1}(\vec{Y}_i - \vec{\mu}_i))\Big)d\vec{B}_i \\
&= (2\pi)^{-(l+N)/2}(\det|\vec{\tau}_i I|)^{-1/2}(\det|\vec{\gamma_1} I|)^{-1/2}\exp\Big(-\frac{1}{2}((\vec{Y}_i - \vec{\mu}_i)^T(\vec{\tau}_i I)^{-1}(\vec{Y}_i - \vec{\mu}_i) \\
&+ \frac{1}{2}(V^{-1}D_i^T(\vec{\tau}_i I)^{-1}(\vec{Y}_i - \vec{\mu}_i))^T V(V^{-1}D_i^T(\vec{\tau}_i I)^{-1}(\vec{Y}_i - \vec{\mu}_i)))\Big)(2\pi)^{l/2}(\det|V|)^{1/2} \\
&= (2\pi)^{-N/2}\gamma_1^{-l/2}\Big(\prod_{q=1}^{N}\tau_{ik_{(q)}}^{-1/2}\Big)(\det|V^{-1}|)^{1/2}\exp\Big(-\frac{1}{2}(\vec{Y}_i - \vec{\mu}_i)^T(\vec{\tau}_i I)^{-1}(\vec{Y}_i - \vec{\mu}_i) \\
&+ \frac{1}{2}(V_i^{-1}(D_i)^T(\vec{\tau}_i I)^{-1}(\vec{Y}_i - \vec{\mu}_i))^T V(V^{-1}(D_i^T(\vec{\tau}_i I)^{-1}(\vec{Y}_i - \vec{\mu}_i)))\Big) \\
&= (2\pi)^{-N/2}\gamma_1^{-l/2}\Big(\prod_{q=1}^{N}\tau_{ik_{(q)}}^{-1/2}\Big)(\det|V|)^{1/2}\exp\Big(-\frac{1}{2}(\vec{Y}_i - \vec{\mu}_i)^T(\vec{\tau}_i I)^{-1}(\vec{Y}_i - \vec{\mu}_i) \\
&+ \frac{1}{2}((D_i)^T(\vec{\tau}_i I)^{-1}(\vec{Y}_i - \vec{\mu}_i))^T(V^{-1})(D_i^T(\vec{\tau}_i I)^{-1}(\vec{Y}_i - \vec{\mu}_i)))\Big).
\end{aligned}
$$

Then, we can have the following form

$$\mathbf{I}^i_{j_1,\ldots,j_l} = (2\pi)^{-N/2}\gamma_1^{-l/2}\left(\prod_{q=1}^{N}\tau_{ik_{(q)}}^{-1/2}\right)(\det|V^{-1}|)^{1/2}\exp\Big(-\frac{1}{2}(\vec{Y}_i-\vec{\mu}_i)^T\delta_{q\tilde{q}}(\vec{\tau}_i I_{N\times N})^{-1}(\vec{Y}_i-\vec{\mu}_i) + \frac{1}{2}(V^{-1}(D_i^T(\vec{\tau}_i I)^{-1}(\vec{Y}_i-\vec{\mu}_i))^T V(V^{-1}(D_i^T(\vec{\tau}_i I)^{-1}(\vec{Y}_i-\vec{\mu}_i)))\Big).$$

Remark:

We note that we can calculate r and $\tilde{r}$ using $\mathbf{I}^i_{j_1,\ldots,j_i}$. For given $(j_1,\ldots,j_l)\in Pa(i)$ when $E_{ij}=1$ and $j=j_m$, the ratio r can be calculated as follows:

$$\begin{aligned} r &= \frac{\mathbf{I}^i_{j_1,\ldots,j_{l_2}}\text{when } E_{ji}=1 \text{ and } j=j_m}{\mathbf{I}^i_{j_1,\ldots,j_{l_2}}\text{when } E_{ji}=0 \text{ and } j=j_m} \\ &= \frac{\mathbf{I}^i_{j_1,\ldots,j_{m-1},j_m,j_{m+1},\ldots,j_{l_2}}}{\mathbf{I}^i_{j_1,\ldots,j_{m-1},j_{m+1},\ldots,j_l}}. \end{aligned} \tag{r}$$

For given $(j_1,\ldots,j_{l_1})\in Pa(i)$ when $(E_{ij}=1 \text{ and } j=j_{m_1})$ and $(i_1,\ldots,i_{l_2})\in Pa(j)$ when $(E_{ji}=1 \text{ and } i=i_{m_2})$, $\tilde{r}$ can be calculated as the following:

$$\tilde{r} = \frac{\mathbf{I}^i_{j_1,\ldots,j_{m_1-1},j_{m_1},j_{m_1+1},\ldots,j_{l_1}}\times\mathbf{I}^j_{i_1\ldots,i_{m_2-1},i_{m_2+1},\ldots,i_{l_2}}}{\mathbf{I}^i_{j_1,\ldots,j_{m_1-1},j_{m_1+1},\ldots,j_{l_1}}\times\mathbf{I}^j_{i_1\ldots,i_{m_2-1},i_{m_2},i_{m_2+1},\ldots,i_{l_2}}}. \tag{$\tilde{r}$}$$

B.10 MH enhanced with Two-Phase Simulated Annealing

To improve the MCMC sample for E_{ij} in MH, which was stuck in local maxima due to the binary value of E_{ij}, we enhance MH using two-phase simulated annealing.

We first define $\eta(m_c)$, a function of the m_cth MCMC iteration, which has the following properties.

Properties of $\eta(m_c)$

- $\eta(0)=1$;
- $\eta(\frac{N_I}{2})=0$, where N_I is the total number of iterations at MCMC;
- $\eta(m_c)$ is decreasing in m_c.

Such a function can be chosen as $\eta(m_c) = e^{-c_1 \frac{m_c}{N_I}}$ that satisfies these properties. It is decreasing and $\eta(0) = 1$. We choose c_1 so that $\eta(\frac{N_I}{2}) \approx 0$.

So, next, we define the following quantity by adapting the idea of simulated annealing:

$$\begin{aligned}\mathbf{I}^{\mathbf{i(m_c)}}_{\mathbf{j_1},\ldots,\mathbf{j_l}} &= \mathbf{I}^{i}_{j_1,\ldots,j_l}\left((2\pi)^{-(l+G)}(\det|\vec{\tau}_i I|)^{-1/2}(\det|\Sigma|)^{-1/2}\mathbf{I}^{i}_{j_1,\ldots,j_l}\right)^{\eta(m_c)} \\ &= \Big(\underbrace{(2\pi)^{-(l+G)}}_{\text{Ⓐ}}\underbrace{(\det|\vec{\tau}_i I|)^{1/2}(\det|\Sigma|)^{1/2}}_{\text{Ⓑ}}\Big)^{\eta(m_c)}\left(\mathbf{I}^{i}_{j_1,\ldots,j_l}\right)^{1+\eta(m_c)},\end{aligned}$$

where

- Ⓐ $= (2\pi)^{-(l+G)}$ is a normalizing constant factor before the integration w.r.t. μ;
- Ⓑ $= (\det|\vec{\tau}_i I|)^{1/2}(\det|\Sigma|)^{1/2}$ is a normalizing constant factor in $\mathbf{I}^{i}_{j_1,\ldots,j_l}$.

We can easily observe that

$$\mathbf{I}^{\mathbf{i(m_c)}}_{\mathbf{j_1},\ldots,\mathbf{j_l}} \approx \mathbf{I}^{\mathbf{i}}_{\mathbf{j_1},\ldots,\mathbf{j_l}}$$

for $\frac{N_I}{2} \le m_c \le N_I$ since $\eta(\frac{N_I}{2}) \approx 10^{-8} \approx 0$.

For updating E_{ij} being 1 or 0, we consider M-H algorithm using this relation between $\mathbf{I}^{i(m_c)}$ and $\mathbf{I}^{i}$:

$$r = \frac{[Y_1,...,Y_p|E_{21},\ldots,E_{ij}=1,...,E_{pp-1}]}{[Y_1,...,Y_p|E_{21},\ldots,E_{ij}=0,...,E_{pp-1}]}$$

and

$$\begin{aligned} r_{m_c} &= \frac{\mathbf{I}^{\mathbf{i(m_c)}}_{\mathbf{j_1},\ldots,\mathbf{j_l}} \text{ when } E_{ij}=1}{\mathbf{I}^{\mathbf{i(m_c)}}_{\mathbf{j_1},\ldots,\mathbf{j_l}} \text{ when } E_{ij}=0} \\ &= \left(\frac{\gamma_1}{2\pi}\right)^{\eta(m_c)}\left(\frac{\mathbf{I}^{\mathbf{i}}_{\mathbf{j_1},\ldots,\mathbf{j_l}} \text{ when } E_{ij}=1}{\mathbf{I}^{\mathbf{i}}_{\mathbf{j_1},\ldots,\mathbf{j_l}} \text{ when } E_{ij}=0}\right)^{1+\eta(m_c)} \\ &= \left(\frac{\gamma_1}{2\pi}\right)^{\eta(m_c)}\left(\frac{[Y_1,...,Y_p|E_{21},\ldots,E_{ij}=1,...,E_{pp-1}]}{[Y_1,...,Y_p|E_{21},\ldots,E_{ij}=0,...,E_{pp-1}]}\right)^{1+\eta(m_c)} \\ &= \left(\frac{\gamma_1}{2\pi}\right)^{\eta(m_c)}(r)^{1+\eta(m_c)}. \end{aligned}$$

As a result, we can have the following facts:

$$r_{m_c} = \begin{cases} (\frac{\gamma_1}{2\pi})^{\eta(m_c)}(r)^{1+\eta(m_c)}, & \text{for } m_c < \frac{N_I}{2}, \\ r, & \text{for } m_c \geq \frac{N_I}{2}. \end{cases}$$

Interpretation of this function can be explained in term of two phases as described as follows;

- Phase 1 (adaptive phase): Acceptance ratio evolves according to $\eta(m_c)$, allowing either stricter or more exploratory behavior depending on $\gamma_1/(2\pi)$.
- Phase 2 (frozen phase): Acceptance ratio is fixed at r, producing a randomized but stable search.

We have an adaptive phase until half of the total MCMC samples. After that, we have the frozen phase. If the value of $r_{m_c} \geq 1$, then we consider that there is a node from $i \to j$. If there is no circle when $E_{ij} = 1$, then accept $E_{ij} = 1$; $E_{ji} = 1$ otherwise. If less than 1, there is a probability r_{m_c} that there is a node from $i \to j$. If $E_{ij} = 0$ with probability r_{m_c}, then accept $E_{ij} = 0$.

Next, if $E_{ij} = 1$ in the previous step, we then update $(E_{ij}, E_{ji}) = (1, 0)$ or (0,1). We define $\tilde{r}_{m_c}$ as

$$\tilde{r}_{m_c} = \begin{cases} (\tilde{r})^{1+\eta(m_c)}, & \text{for } m_c < \frac{N_I}{2}, \\ \tilde{r}, & \text{for } m_c \geq \frac{N_I}{2}. \end{cases}$$

If the value $\tilde{r}_{m_c} > 1$, then we accept that there is a node from $i \to j$ but not $j \to i$, that is $(E_{ij}, E_{ji}) = (1, 0)$. If less than 1, there is a probability $\tilde{r}_{m_c}$ that there is a node form $j \to i$ but not $i \to j$, that is $(E_{ij}, E_{ji}) = (0, 1)$.

C Derivation of Full Conditional Distribution Under DCG

The full conditional distributions for all parameters except (E, B) are identical to those provided in Appendix B. We therefore proceed to detail the sampling procedure from joint full conditional distribution $[E, B|-]$ in this section.

C.1 Adjacency and Causal Effect Matrix Update

The joint full conditional distribution of $[B, E|-]$ is

$$[B_{12}, \ldots, B_{pp-1}, E_{12}, \ldots, E_{pp-1}|-] \propto [\vec{Y}_1, \ldots, \vec{Y}_p|B_{12}, \ldots, B_{pp-1}, \vec{\mu}_1, \ldots, \vec{\mu}_p, \vec{\tau}_1, \ldots, \vec{\tau}_p]$$
$$[B_{12}|E_{12}, \gamma_1] \cdots [B_{pp-1}|E_{pp-1}, \gamma_1][E_{12}|\gamma] \cdots [E_{pp-1}|\gamma].$$

To update E_{ij} and B_{ij}, we assume that $E_{ij} = 1$ and from a proposal $v \sim N(0, 1)$, $B_{ij} = v$, then for a Metropolis Hastings ratio

$$r = \frac{[B_{12}, \ldots, B_{ij} = v, \ldots, B_{pp-1}, E_{12}, \ldots, E_{ij} = 1, \ldots, E_{pp-1}|-]}{[B_{12}, \ldots, B_{ij} = 0, \ldots, B_{pp-1}, E_{12}, \ldots, E_{ij} = 0, \ldots, E_{pp-1}|-]}.$$

Hence, the numerator of r can have the following expression:

$$[\vec{Y}_i|B_{12}, \ldots, B_{ij} = v, \ldots, B_{jp}, \vec{Y}_1, \ldots, \vec{Y}_{i-1}, \vec{Y}_{i+1}, \ldots \vec{Y}_p][B_{12}|E_{12}, \gamma_1][E_{12}|\gamma] \cdots [B_{ij} = v|E_{ij}, \gamma_1]$$
$$[E_{ij} = 1|\gamma] \cdots [B_{pp-1}|E_{pp-1}, \gamma_1][E_{pp-1}|\gamma]$$
$$= [D_i\vec{B}_i + MVN(\vec{\mu}_i, \vec{\tau}_i I)][N(B_{12}; 0, \gamma_1)] \cdots [N(B_{ij} = v; 0, \gamma_1)] \cdots [N(B_{pp-1}; 0, \gamma_1)]$$
$$\times \gamma^{k+1}(1-\gamma)^{(p(p-1))-(k+1)}.$$

C.2 Metropolis Hastings Adjacency and Causal Update

For the cyclic sampler, each ordered pair (i, j) is updated via a joint Metropolis–Hastings birth/death move on (E_{ij}, B_{ij}). We use the spike–and–slab proposal kernel

$$q(E'_{ij}, B'_{ij} \mid E_{ij}, B_{ij}) = \mathbb{I}(E_{ij} = 0)\,\mathbb{I}(E'_{ij} = 1)\,\phi(B'_{ij}; 0, \sqrt{\gamma_1}) \;+\; \mathbb{I}(E_{ij} = 1)\,\mathbb{I}(E'_{ij} = 0)\,\delta_0(B'_{ij}),$$

where (E_{ij}, B_{ij}) denotes the current state and (E'_{ij}, B'_{ij}) the proposed state. In particular, if $E_{ij} = 0$, we propose adding the edge by drawing $B'_{ij} \sim N(0, \gamma_1)$. If $E_{ij} = 1$, we propose deleting the edge by setting $B'_{ij} = 0$.

C.2.1 Removing an edge $E_{ij} = 1 \to E'_{ij} = 0$

Let $P(B_{ij}, E_{ij} \mid -)$ denote the (unnormalized) full conditional of (B_{ij}, E_{ij}). Starting from a present edge $(E_{ij}, B_{ij}) = (1, B_{ij})$, we propose deletion by setting $(E'_{ij}, B'_{ij}) = (0, 0)$. The Metropolis–Hastings ratio is

$$r = \frac{P(B'_{ij}, E'_{ij} \mid -)}{P(B_{ij}, E_{ij} \mid -)} \cdot \frac{q(E_{ij}, B_{ij} \mid E'_{ij}, B'_{ij})}{q(E'_{ij}, B'_{ij} \mid E_{ij}, B_{ij})}.$$

Under the birth/death proposal, the forward proposal is the following,

$$q(E'_{ij}, B'_{ij} \mid E_{ij}, B_{ij}) = \mathbb{I}(E'_{ij} = 0)\,\delta_0(B'_{ij}),$$

while the reverse proposal corresponds to re-adding the edge with a slab draw,

$$q(E_{ij}, B_{ij} \mid E'_{ij}, B'_{ij}) = \mathbb{I}(E_{ij} = 1)\,\phi(B_{ij}; 0, \sqrt{\gamma_1}).$$

Hence, the proposal ratio contributes the factor $\phi(B_{ij}; 0, \sqrt{\gamma_1})$.

C.2.2 Adding an edge $E_{ij} = 0 \to E'_{ij} = 1$

Starting from an absent edge $(E_{ij}, B_{ij}) = (0, 0)$, we propose addition by drawing $B'_{ij} \sim N(0, \gamma_1)$ and setting $E'_{ij} = 1$. The Metropolis–Hastings ratio is

$$r = \frac{P(B'_{ij}, E'_{ij} \mid -)}{P(B_{ij}, E_{ij} \mid -)} \cdot \frac{q(E_{ij}, B_{ij} \mid E'_{ij}, B'_{ij})}{q(E'_{ij}, B'_{ij} \mid E_{ij}, B_{ij})}.$$

The forward proposal is the slab density,

$$q(E'_{ij}, B'_{ij} \mid E_{ij}, B_{ij}) = \mathbb{I}(E'_{ij} = 1)\, \phi(B'_{ij}; 0, \sqrt{\gamma_1}),$$

and the reverse proposal is deterministic deletion back to zero,

$$q(E_{ij}, B_{ij} \mid E'_{ij}, B'_{ij}) = \mathbb{I}(E_{ij} = 0)\, \delta_0(B_{ij}).$$

Thus, the proposal ratio contributes the factor $1/\phi(B'_{ij}; 0, \sqrt{\gamma_1})$.

C.2.3 Adaptive proposal variance for causal effect updates

For cyclic graphs, proposals that add an edge require sampling B'_{ij}. To stabilize mixing we create the following adaptive rule:

- During burn in (first 1000 iterations), we use a fixed proposal standard deviation at $\sigma_{add} = 0.15$.
- After burn in, the proposal standard deviation matches the slab scale at $\sigma_{add} = \sqrt{\gamma_1}$.

Therefore, the proposal is

$$B'_{ij} \sim N(0, \sigma^2_{add}). \tag{2}$$

The purpose of the adaptive proposal variance serves two purposes:

(i) The fixed 0.15 standard deviation encourages broad exploration and prevents early sticking.

(ii) After burn-in, matching σ_{add} to $\sqrt{\gamma_1}$ adapts the proposal to the posterior scale, improving mixing and acceptance rates.

Due to the more complex geometry of the posterior space of cyclic graphs, this ensures more robust exploration of both the adjacency and causal spaces.

C.3 Random Walk Update of Causal Weights

C.3.1 Adaptive random walk proposal variance schedule

To improve mixing across the MCMC run, we use an adaptive proposal variance that gradually increases from a small exploratory scale to a larger, posterior aligned scale. Let t denote the MCMC iteration. For the random walk proposal the standard deviation σ_{rw}^2 is defined as,

$$\sigma_{rw}(t) = 0.03 + 0.07\Big(\frac{min(t, 15000)}{15000}\Big). \tag{3}$$

This rule has the following three interpretation:

(i) **Early iterations (exploration):**

When t is less than 15000, the proposal scale is close to 0.03, producing small, conservative random walk steps that stabilize the chain while it adapts to the posterior geometry.

(ii) **Later iterations (posterior sensitive updates):**

As t increases, the proposal scale grows linearly to 0.10, encouraging larger moves and escaping local modes.

(iii) **Clipping at iteration 15000:**

This caps the adaptation after iteration 15000, preventing uncontrolled growth and maintaining a stable long run proposal distribution.

This adaptive random walk mechanism complements the spike and slab edge proposals by allowing flexible local exploration of the causal effect posterior, which is especially important for DCGs.

C.4 Projection to the Disjoint Cycle Space

At the end of the unconstrained burn-in phase, the current state (E, B) is mapped to the admissible disjoint-cycle graph space before the constrained sampling phase begins. Rather than explicitly enumerating all directed cycles, we first partition the graph into strongly connected components (SCCs) using Tarjan's algorithm from [Tarjan, 1972].

A strongly connected component is a maximal set of nodes that are mutually reachable. More precisely, a set $\mathcal{S} \subseteq \{1, \ldots, p\}$ is an SCC if, for every pair $i, j \in \mathcal{S}$, there exists a directed path from i to j and also a directed path from j back to i. Tarjan's algorithm partitions the graph into disjoint SCCs.

A singleton SCC contains only one node and is trivial in the absence of self-loops. In contrast, an SCC containing at least two nodes is nontrivial and necessarily contains at least one directed cycle. For example,

$$1 \to 2 \to 3 \to 1$$

forms the nontrivial SCC $\{1, 2, 3\}$. If two directed cycles share a node, their nodes belong to the same SCC. For example,

$$1 \to 2 \to 3 \to 1, \qquad 2 \to 4 \to 2$$

produce the single SCC $\{1, 2, 3, 4\}$. Thus, SCCs provide a convenient way to identify portions of the graph in which overlapping directed cycles may occur.

For each nontrivial SCC $\mathcal{S}$, we examine only the edges whose endpoints both belong to $\mathcal{S}$. Let

$$E_{\mathcal{S}} = (E_{ij} : i, j \in \mathcal{S})$$

denote the induced adjacency matrix. Under our convention, $E_{ij} = 1$ represents the directed edge $j \to i$. Therefore, the internal in-degree and out-degree of node $i \in \mathcal{S}$ are

$$d_i^{\text{in}}(\mathcal{S}) = \sum_{j \in \mathcal{S}} E_{ij}, \qquad d_i^{\text{out}}(\mathcal{S}) = \sum_{j \in \mathcal{S}} E_{ji}.$$

A nontrivial SCC forms a single simple directed cycle if and only if every node has exactly one incoming edge and one outgoing edge within that SCC:

$$d_i^{\text{in}}(\mathcal{S}) = d_i^{\text{out}}(\mathcal{S}) = 1, \qquad \forall i \in \mathcal{S}.$$

For example, in

$$1 \to 2 \to 3 \to 1,$$

each node has exactly one incoming and one outgoing edge, so the SCC is a single simple directed cycle. In contrast, if node 2 also belongs to the cycle $2 \to 4 \to 2$, then the resulting SCC contains a node with more than one incoming or outgoing edge and therefore contains overlapping cycles. Consequently, the disjoint-cycle restriction is satisfied when every nontrivial SCC is a single simple directed cycle.

If a nontrivial SCC violates this condition, we identify internal edges incident to nodes responsible for the degree violation. Since $E_{ij} = 1$ corresponds to $j \to i$, the candidate edge set is

$$\mathcal{C}_{\mathcal{S}} = \left\{(i,j) : E_{ij} = 1,\ i,j \in \mathcal{S},\ d_i^{\text{in}}(\mathcal{S}) > 1 \text{ or } d_j^{\text{out}}(\mathcal{S}) > 1\right\}.$$

Thus, an edge $j \to i$ is considered for removal when its child i has more than one incoming edge within the SCC or its parent j has more than one outgoing edge within the SCC.

Among these candidate edges, we remove the edge having the smallest absolute causal effect:

$$(i^*, j^*) = \arg\min_{(i,j)\in\mathcal{C}_{\mathcal{S}}} |B_{ij}|.$$

The corresponding adjacency and causal-effect entries are then set to zero,

$$E_{i^*j^*} = 0, \qquad B_{i^*j^*} = 0.$$

Removing an edge can change the SCC decomposition of the graph. Therefore, after each removal, the SCCs are recomputed and the procedure is repeated. The projection terminates when every nontrivial SCC satisfies the one-incoming-edge and one-outgoing-edge condition and therefore represents a single simple directed cycle.

The resulting pair (E, B) provides the initial state for the constrained post-burn-in sampler. Stability is enforced separately during the constrained sampling phase through the spectral-radius condition

$$\rho(B) < 1.$$

Proposals that violate this condition are rejected rather than rescaling the causal effect matrix.

Algorithm 3 Projection to the Disjoint Cycle Space

Input: Current burn-in state (E, B)

Set $\mathrm{diag}(E) = \mathrm{diag}(B) = 0$

Set $B_{ij} = 0$ whenever $E_{ij} = 0$

while E does not satisfy the disjoint cycle restriction **do**

    Compute the strongly connected components of E

    Select a nontrivial SCC $\mathcal{S}$ that is not a simple directed cycle

    Compute

$$d_i^{\text{in}}(\mathcal{S}) = \sum_{j \in \mathcal{S}} E_{ij}, \qquad d_i^{\text{out}}(\mathcal{S}) = \sum_{j \in \mathcal{S}} E_{ji}$$

    for all $i \in \mathcal{S}$

    Define

$$\mathcal{C}_{\mathcal{S}} = \left\{(i,j) : E_{ij} = 1,\ i,j \in \mathcal{S},\ d_i^{\text{in}}(\mathcal{S}) > 1 \text{ or } d_j^{\text{out}}(\mathcal{S}) > 1\right\}$$

    Select

$$(i^*, j^*) = \arg\min_{(i,j) \in \mathcal{C}_{\mathcal{S}}} |B_{ij}|$$

    Set $E_{i^*j^*} = 0$ and $B_{i^*j^*} = 0$

end while

Return: projected (E, B)

D Consistency of the SHD Bayes Action

The following result describes the behavior of the SHD-based decision-theoretic graph summary when the posterior concentrates on the true graph.

Let $\mathcal{D}$ be the finite admissible graph space for fixed p, and suppose that the true graph $\mathcal{G}_0$ belongs to $\mathcal{D}$. If

$$\Pi(\mathcal{G}_0 \mid \vec{Y}) \to 1$$

as the sample size tends to infinity, then the Bayes action under Structural Hamming Distance,

$$\widehat{\mathcal{G}} = \arg\min_{a \in \mathcal{D}} \sum_{g \in \mathcal{D}} d_{\mathrm{SHD}}(a, g)\Pi(g \mid \vec{Y}),$$

converges to $\mathcal{G}_0$.

Proof. Define the posterior expected SHD of choosing graph a as

$$R(a) = \sum_{g \in \mathcal{D}} d_{\mathrm{SHD}}(a, g)\Pi(g \mid \vec{Y}).$$

Because $\mathcal{D}$ is finite for fixed p, define

$$D_{\max} = \max_{g \in \mathcal{D}} d_{\mathrm{SHD}}(\mathcal{G}_0, g) < \infty.$$

For the true graph,

$$\begin{aligned} R(\mathcal{G}_0) &= \sum_{g \neq \mathcal{G}_0} d_{\mathrm{SHD}}(\mathcal{G}_0, g)\Pi(g \mid \vec{Y}) \\ &\leq D_{\max} \sum_{g \neq \mathcal{G}_0} \Pi(g \mid \vec{Y}) \\ &= D_{\max}\{1 - \Pi(\mathcal{G}_0 \mid \vec{Y})\}. \end{aligned}$$

Hence,

$$R(\mathcal{G}_0) \to 0.$$

Now consider any $a \neq \mathcal{G}_0$. Since every term in $R(a)$ is nonnegative,

$$R(a) \geq d_{\mathrm{SHD}}(a, \mathcal{G}_0)\Pi(\mathcal{G}_0 \mid \vec{Y}).$$

Distinct graphs differ by at least one edge operation, so

$$d_{\mathrm{SHD}}(a, \mathcal{G}_0) \geq 1,$$

and therefore

$$R(a) \geq \Pi(\mathcal{G}_0 \mid \vec{Y}).$$

Writing $\Pi_0 = \Pi(\mathcal{G}_0 \mid \vec{Y})$, we therefore have

$$R(\mathcal{G}_0) \leq D_{\max}(1 - \Pi_0), \qquad R(a) \geq \Pi_0$$

for every $a \neq \mathcal{G}_0$. Whenever

$$\Pi_0 > \frac{D_{\max}}{D_{\max} + 1},$$

it follows that

$$R(\mathcal{G}_0) < R(a) \qquad \text{for every } a \neq \mathcal{G}_0.$$

Since $\Pi_0 \to 1$, this condition holds for sufficiently large sample size. Thus $\mathcal{G}_0$ is eventually the unique minimizer of posterior expected SHD, and

$$\widehat{\mathcal{G}} \to \mathcal{G}_0.$$

□

E Simulation Results Table

E.1 Acyclic Simulation Tables

		$p = 20$			$p = 30$			$p = 40$		
N	Metric	ICA-L	Direct-L	`BayesDAG`	ICA-L	Direct-L	`BayesDAG`	ICA-L	Direct-L	`BayesDAG`
	TPR	0.0913	0.5311	0.9748	0.3245	0.8656	0.9524	0.3538	0.7639	0.9602
	FPR	0.0086	0.0128	0.0104	0.0282	0.0517	0.0122	0.0498	0.1120	0.0127
	Acc	0.9059	0.9543	0.9882	0.9092	0.9403	0.9844	0.8920	0.8759	0.9846
100	Precision	0.5326	0.7930	0.9105	0.5534	0.8943	0.8873	0.4350	0.4350	0.8918
	Recall	0.0913	0.5311	0.9748	0.3245	0.8656	0.9524	0.3538	0.7639	0.9602
	F1	0.1603	0.6295	0.9410	0.9414	0.7450	0.9221	0.3895	0.5525	0.9245
	Time	0.0831	1.025	362.0	0.1642	2.499	825.4	0.1946	3.228	1735
	TPR	0.4934	0.9987	0.9579	0.3297	0.9946	0.9845	0.3992	0.9937	0.9681
	FPR	0.0229	0.00597	0.0105	0.0258	0.0098	0.0073	0.0496	0.0123	0.0128
	Acc	0.9312	0.9945	0.9865	0.9119	0.9906	0.9919	0.8965	0.9883	0.9853
200	Precision	0.7040	0.9481	0.9105	0.5787	0.9372	0.9390	0.4646	0.8996	0.8938
	Recall	0.4934	0.9987	0.9579	0.3297	0.9946	0.9845	0.3992	0.9937	0.9681
	F1	0.5783	0.9723	0.9329	0.4189	0.9538	0.9600	0.4286	0.9438	0.9290
	Time	0.1058	1.151	625.9	0.2371	2.677	1483	0.2316	3.339	3177
	TPR	0.9353	1	0.9593	0.3993	0.9998	0.9836	0.3451	0.9937	0.9537
	FPR	0.00939	0.00376	0.0114	0.0430	0.0050	0.0083	0.0506	0.0479	0.0199
	Acc	0.9853	0.9966	0.9858	0.9031	0.9955	0.9909	0.8909	0.9555	0.9775
300	Precision	0.9207	0.9662	0.9075	0.5010	0.9304	0.9227	0.4245	0.7243	0.8448
	Recall	0.9353	1	0.9593	0.3993	0.9998	0.9836	0.3451	0.9937	0.9537
	F1	0.9266	0.9826	0.9315	0.4436	0.9772	0.9556	0.3798	0.8282	0.9537
	Time	0.0336	0.8074	866.9	0.1879	1.932	2109	0.2719	3.584	4585

Table 2: Performance comparison among three methods (ICALiNGAM, DirectLiNGAM, `BayesDAG`) in terms TPR, FPR, Acc, Precision, Recall, and F1: 100 simulated data sets on simulation setting; the error terms $\epsilon_i^{(q)} \sim \sum_{k=1}^{M_{true}} \pi_{ik} N(\mu_{ik}, \tau_{ik})$, where $M_{true} = 2$, $(\mu_{i1}, \mu_{i2}) = (-0.5, 0.5)$, and $(\tau_{i1}, \tau_{i2}) = (0.1, 0.3)$, and $(\pi_{i1}, \pi_{i2}) = (0.5, 0.5)$, $B = E$, $(\Delta, 1-\Delta)$=(0.9,0.1); Average value of each metric is summarized in this table; We fit our methods with a misspecified mixture $M_{mis} = 5$.

		$p = 20$			$p = 30$			$p = 40$		
N	Metric	ICA-L	Direct-L	`BayesDAG`	ICA-L	Direct-L	`BayesDAG`	ICA-L	Direct-L	`BayesDAG`
	TPR	0.2492	0.7403	0.9087	0.2841	0.6837	0.9269	0.3238	0.6142	0.9561
	FPR	0.0203	0.0434	0.0250	0.0281	0.06852	0.0221	0.0499	0.1294	0.0184
	Acc	0.9103	0.9360	0.9687	0.9054	0.9075	0.9730	0.8890	0.8456	0.9791
100	Precision	0.5690	0.6553	0.7956	0.5201	0.5275	0.8197	0.4125	0.3463	0.8496
	Recall	0.2492	0.7403	0.9087	0.3937	0.6837	0.9269	0.3238	0.6142	0.9561
	F1	0.3436	0.7403	0.8473	0.3664	0.5938	0.8695	0.3621	0.4418	0.8995
	Time	0.0867	1.356	362.0	0.1644	2.821	831.3	0.2976	5.627	1828
	TPR	0.3421	0.9034	0.9553	0.2770	0.8733	0.9732	0.3421	0.8346	0.9765
	FPR	0.0343	0.0160	0.0195	0.0258	0.0346	0.0189	0.0400	0.0708	0.0187
	Acc	0.9065	0.9763	0.9781	0.9067	0.9564	0.9803	0.8998	0.9199	0.9809
200	Precision	0.5145	0.8658	0.8417	0.5366	0.7453	0.8488	0.4807	0.5838	0.8524
	Recall	0.3421	0.9034	0.9553	0.2770	0.8733	0.9732	0.3421	0.8346	0.9765
	F1	0.4083	0.8830	0.8940	0.3642	0.8018	0.9063	0.3989	0.6825	0.9765
	Time	0.1107	1.533	632.6	0.2681	3.804	1538	0.3595	6.237	3162
	TPR	0.7511	0.9716	0.9581	0.3387	0.9474	0.9533	0.2871	0.9397	0.9642
	FPR	0.0428	0.0060	0.0200	0.0427	0.0148	0.0246	0.0437	0.0278	0.0249
	Acc	0.9376	0.9919	0.9779	0.8975	0.9815	0.9733	0.8911	0.9569	0.9741
300	Precision	0.6701	0.9480	0.8416	0.4638	0.8851	0.8099	0.4173	0.8115	0.8131
	Recall	0.7511	0.9716	0.9581	0.3387	0.9474	0.9533	0.2871	0.9397	0.9642
	F1	0.7052	0.9591	0.8948	0.3904	0.9134	0.8750	0.3397	0.8664	0.8811
	Time	0.1326	1.460	881.4	0.2762	3.447	2186	0.2835	4.136	4805

Table 3: Performance comparison among three methods (ICALiNGAM, DirectLiNGAM, `BayesDAG`) in terms TPR, FPR, Acc, Precision, Recall, and F1: 100 simulated data sets on simulation setting; the error terms $\epsilon_i^{(q)} \sim Laplace(0, 0.25)$, $B = E$, $(\Delta, 1-\Delta)$=(0.9,0.1); Average value of each metric is summarized in this table; We fit our methods with a misspecified mixture $M_{mis} = 5$.

		$p = 20$			$p = 30$			$p = 40$		
N	Metric	ICA-L	Direct-L	`BayesDAG`	ICA-L	Direct-L	`BayesDAG`	ICA-L	Direct-L	`BayesDAG`
	TPR	0.2589	0.5539	0.7868	0.2980	0.5540	0.8166	0.3341	0.5069	0.8580
	FPR	0.0232	0.0860	0.0496	0.0291	0.1042	0.0475	0.0496	0.1612	0.0406
	Acc	0.9086	0.8798	0.9348	0.9058	0.8627	0.9393	0.8903	0.8064	0.9495
100	Precision	0.5461	0.4173	0.6325	0.5235	0.3680	0.6505	0.4228	0.2564	0.6978
	Recall	0.2589	0.5539	0.7868	0.2980	0.5540	0.8166	0.3341	0.5069	0.8580
	F1	0.3475	0.4735	0.6998	0.3786	0.4412	0.7235	0.3727	0.3402	0.8580
	Time	0.1108	1.793	382.7	0.2096	3.498	905.2	0.2290	4.322	1806
	TPR	0.3429	0.7456	0.8086	0.3005	0.6941	0.8437	0.3726	0.6308	0.8254
	FPR	0.0395	0.0554	0.0545	0.0268	0.0918	0.0573	0.0400	0.1555	0.0716
	Acc	0.9018	0.9295	0.9325	0.9082	0.8875	0.9331	0.9028	0.8236	0.9184
200	Precision	0.4800	0.5527	0.6185	0.5467	0.4624	0.6169	0.5019	0.3113	0.5600
	Recall	0.3429	0.7456	0.8086	0.3005	0.6941	0.8437	0.3726	0.6308	0.8254
	F1	0.3984	0.6301	0.6989	0.3865	0.5528	0.7118	0.4272	0.4158	0.6663
	Time	0.1289	1.793	672.9	0.3062	4.331	1655	0.3577	6.191	3578
	TPR	0.5905	0.7966	0.7990	0.3552	0.7618	0.8217	0.3067	0.7141	0.7894
	FPR	0.0710	0.0488	0.0677	0.0437	0.0761	0.0752	0.0425	0.1354	0.1054
	Acc	0.8969	0.9365	0.9196	0.8982	0.9077	0.9149	0.8941	0.8499	0.8844
300	Precision	0.4719	0.4399	0.5629	0.4673	0.5332	0.5459	0.4277	0.3758	0.4537
	Recall	0.5905	0.7966	0.7990	0.3552	0.7618	0.8217	0.3067	0.7141	0.7894
	F1	0.5233	0.7129	0.6585	0.4028	0.6612	0.6546	0.3609	0.4902	0.5750
	Time	0.1980	1.840	1392	0.2739	3.539	2448	0.4283	6.602	5661

Table 4: Performance comparison among three methods (ICALiNGAM, DirectLiNGAM, `BayesDAG`) in terms TPR, FPR, Acc, Precision, Recall, and F1: 100 simulated data sets on simulation setting; the error terms $\epsilon_i^{(q)} \sim t_{df=7}$, $B = E$, $(\Delta, 1-\Delta)$=(0.9,0.1); Average value of each metric is summarized in this table; We fit our methods with a misspecified mixture $M_{mis} = 5$.

E.2 Cyclic Simulation Tables

		$p = 20$			$p = 30$			$p = 40$		
N	Metric	LiNG	`MR.RGM`	`BayesDCG`	LiNG	`MR.RGM`	`BayesDCG`	LiNG	`MR.RGM`	`BayesDCG`
	TPR	0.9487	0.9947	0.8656	0.9493	0.9929	0.7977	0.9435	0.9955	0.7642
	FPR	0.9155	0.9402	0.0216	0.9103	0.9573	0.0277	0.9051	0.9660	0.0335
	Acc	0.1709	0.1487	0.9677	0.1756	0.1345	0.9555	0.1797	0.1278	0.9468
100	Precision	0.1033	0.1000	0.8119	0.1038	0.1000	0.7566	0.1038	0.1002	0.7125
	Recall	0.9487	0.9947	0.8656	0.9493	0.9929	0.7977	0.9435	0.9955	0.7642
	F1	0.1863	0.1817	0.8369	0.1872	0.1815	0.7761	0.1870	0.1820	0.7371
	Time	0.2779	27.25	151.7	0.1500	87.87	586.1	0.1834	211.8	2791
	TPR	0.9911	0.9939	0.9492	0.9470	0.9943	0.9271	0.9422	0.9955	0.8763
	FPR	0.5699	0.9385	0.0131	0.9104	0.9567	0.0159	0.9038	0.9660	0.0262
	Acc	0.4862	0.1501	0.9833	0.1753	0.1353	0.9786	0.1808	0.1278	0.9643
200	Precision	0.1694	0.10005	0.8896	0.1036	0.10009	0.8653	0.1038	0.1002	0.7889
	Recall	0.9911	0.9939	0.9491	0.9470	0.9943	0.9271	0.9422	0.9492	0.8763
	F1	0.2880	0.1818	0.9175	0.1868	0.1819	0.8946	0.1870	0.1820	0.8296
	Time	0.1520	27.08	183	0.1801	87.37	450.3	0.2034	211.8	1750.4
	TPR	1.0000	0.9939	0.9532	0.9493	0.9940	0.9308	0.9439	0.9942	0.9101
	FPR	0.3491	0.9386	0.0135	0.8827	0.9558	0.0179	0.9040	0.9637	0.0234
	Acc	0.6858	0.1499	0.9833	0.2005	0.1360	0.9771	0.1808	0.1297	0.9701
300	Precision	0.2434	0.10004	0.8921	0.1068	0.10014	0.8528	0.1040	0.1003	0.8156
	Recall	1.0000	0.9939	0.9532	0.9493	0.9940	0.9308	0.9439	0.9942	0.9101
	F1	0.3910	0.1818	0.9202	0.1920	0.1820	0.8893	0.1873	0.1822	0.8592
	Time	0.1309	27.05	221.5	0.2332	86.84	518.6	0.3062	206.9	1798

Table 5: Performance comparison among LiNG, `MR.RGM`, and `BayesDCG` in terms of TPR, FPR, accuracy, precision, recall, F1 score, and runtime over 100 simulated data sets. The structural errors follow $\epsilon_i^{(q)} \sim \sum_{k=1}^{M_{\text{true}}} \pi_{ik} N(\mu_{ik}, \tau_{ik})$, where $M_{\text{true}} = 2$, $(\mu_{i1}, \mu_{i2}) = (-0.5, 0.5)$, $(\tau_{i1}, \tau_{i2}) = (0.1, 0.3)$, and $(\pi_{i1}, \pi_{i2}) = (0.5, 0.5)$. We use $(\Delta, 1 - \Delta) = (0.9, 0.1)$, and `BayesDCG` is fitted using $M_{\text{mis}} = 5$ mixture components. Average values of each metric are reported.

		$p = 20$			$p = 30$			$p = 40$		
N	Metric	`LiNG`	`MR.RGM`	`BayesDCG`	`LiNG`	`MR.RGM`	`BayesDCG`	`LiNG`	`MR.RGM`	`BayesDCG`
	TPR	0.9547	0.9958	0.7319	0.9561	0.9956	0.7037	0.9462	0.9955	0.7300
	FPR	0.8839	0.9388	0.0507	0.8866	0.9573	0.0465	0.8904	0.9658	0.0425
	Acc	0.2000	0.1499	0.9287	0.1977	0.1348	0.9294	0.1932	0.1280	0.9354
100	Precision	0.1072	0.1002	0.6069	0.1070	0.1002	0.6201	0.1056	0.1002	0.6512
	Recall	0.9547	0.9958	0.7319	0.9561	0.9956	0.7037	0.9462	0.9955	0.7300
	F1	0.1927	0.1821	0.662	0.1925	0.1820	0.6586	0.1900	0.1821	0.6880
	Time	0.2082	26.77	223.2	0.1623	89.37	725	0.3529	212.7	3849
	TPR	0.9766	0.9961	0.8876	0.9602	0.9951	0.8870	0.9508	0.9938	0.8739
	FPR	0.8194	0.9387	0.0323	0.8553	0.9570	0.0319	0.8749	0.9652	0.0364
	Acc	0.2602	0.1501	0.9601	0.2263	0.1350	0.9603	0.2076	0.1283	0.9548
200	Precision	0.1171	0.1002	0.7504	0.1110	0.1001	0.7515	0.1078	0.1001	0.7266
	Recall	0.9766	0.9961	0.8876	0.9602	0.9951	0.8870	0.9508	0.9938	0.8739
	F1	0.2091	0.1821	0.8117	0.1989	0.1819	0.8130	0.1936	0.1819	0.7928
	Time	0.3750	26.55	261.8	0.5236	88.00	783.5	0.2625	210.6	2728
	TPR	0.9950	0.9921	0.9356	0.9743	0.9961	0.9254	0.9588	0.9934	0.9087
	FPR	0.6985	0.9375	0.0236	0.7986	0.9561	0.0271	0.8573	0.9642	0.0345
	Acc	0.3709	0.1509	0.9726	0.2787	0.1359	0.9683	0.2243	0.1292	0.9599
300	Precision	0.1372	0.1000	0.8153	0.1196	0.1003	0.7916	0.1106	0.1002	0.7485
	Recall	0.9950	0.9921	0.9356	0.9743	0.9961	0.9254	0.9588	0.9934	0.9087
	F1	0.2410	0.1817	0.8696	0.2130	0.1823	0.8523	0.1983	0.1820	0.8196
	Time	0.1777	25.99	249.0	0.4963	87.46	646.3	0.2574	208.5	2150

Table 6: Performance comparison among `LiNG`, `MR.RGM`, and `BayesDCG` in terms of TPR, FPR, accuracy, precision, recall, F1 score, and runtime over 100 simulated data sets. The structural errors follow $\epsilon_i^{(q)} \sim \text{Laplace}(0, 0.25)$ with $(\Delta, 1-\Delta) = (0.9, 0.1)$. Average values of each metric are reported. `BayesDCG` is fitted using $M_{\text{mis}} = 5$ mixture components.

		$p = 20$			$p = 30$			$p = 40$		
N	Metric	LiNG	MR.RGM	BayesDCG	LiNG	MR.RGM	BayesDCG	LiNG	MR.RGM	BayesDCG
	TPR	0.9537	0.9958	0.7866	0.9545	0.9946	0.7683	0.9432	0.9958	0.7777
	FPR	0.9008	0.9388	0.0330	0.8945	0.9574	0.0389	0.8955	0.9654	0.0401
	Acc	0.1846	0.1499	0.9499	0.1904	0.1347	0.9425	0.1884	0.1283	0.9421
100	Precision	0.1053	0.1002	0.7197	0.1060	0.1001	0.6813	0.1048	0.1003	0.6790
	Recall	0.9537	0.9958	0.7866	0.9545	0.9946	0.7683	0.9432	0.9958	0.7777
	F1	0.1896	0.1821	0.7503	0.1908	0.1818	0.7216	0.1886	0.1822	0.7246
	Time	0.4629	26.77	173.9	0.2485	86.94	696.6	0.1992	209.5	3009
	TPR	0.9568	0.9961	0.8615	0.9511	0.9951	0.8716	0.9468	0.9942	0.8731
	FPR	0.8920	0.9387	0.0243	0.8863	0.9563	0.0316	0.8887	0.9635	0.0380
	Acc	0.1928	0.1501	0.9649	0.1974	0.1357	0.9590	0.1949	0.1299	0.9533
200	Precision	0.1065	0.1002	0.7932	0.1065	0.1002	0.7505	0.1059	0.1003	0.7181
	Recall	0.9568	0.9961	0.8615	0.9511	0.9951	0.8716	0.9468	0.9942	0.8731
	F1	0.1917	0.1821	0.8248	0.1916	0.1821	0.8059	0.1904	0.1822	0.7873
	Time	0.5915	26.55	207.8	0.2165	86.32	678.4	0.7539	208.3	2681
	TPR	0.9616	0.9921	0.9054	0.9514	0.9941	0.9048	0.9484	0.9935	0.8969
	FPR	0.8769	0.9375	0.0202	0.8766	0.9542	0.0275	0.8841	0.9596	0.0384
	Acc	0.2069	0.1509	0.9728	0.2062	0.1374	0.9660	0.1992	0.1334	0.9553
300	Precision	0.1087	0.1000	0.8347	0.1076	0.1003	0.7839	0.1065	0.1006	0.7245
	Recall	0.9616	0.9921	0.9054	0.9514	0.9941	0.9048	0.9484	0.9935	0.8969
	F1	0.1953	0.1817	0.8671	0.1934	0.1822	0.8393	0.1915	0.1827	0.8001
	Time	0.2407	25.99	290.5	0.2482	84.52	635.2	0.2803	206.3	2488

Table 7: Performance comparison among `LiNG`, `MR.RGM`, and `BayesDCG` in terms of TPR, FPR, accuracy, precision, recall, F1 score, and runtime over 100 simulated data sets. The structural errors follow $\epsilon_i^{(q)} \sim t_{df=7}$ with $(\Delta, 1-\Delta) = (0.9, 0.1)$. Average values of each metric are reported. `BayesDCG` is fitted using $M_{\text{mis}} = 5$ mixture components.